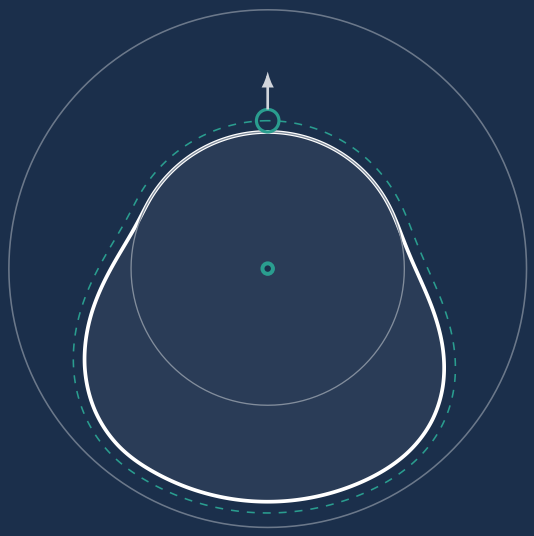
ELECTRONIC CAMMING ON THE TEKNIC CLEARCORE

Electronic CAM Workbench

User Manual

Design, check, and run electronic cam profiles on the Teknic ClearCore with Teknic ClearPath servos



Software version 0.11.4 | ECW-UM-100, Revision C

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Companion document: *Cam Law Mathematics and ClearCore Theory of Operation*, ECW-TOP-100.

November 2026

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Electronic CAM Workbench User Manual
Document ECW-UM-100, Revision C, November 2026
For Electronic CAM Workbench and ECam firmware version 0.11.4

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Source of this manual. The chapters of this manual are the user guide that the Workbench shows with the Help button or the F1 key. Both come from the same files, so they agree. The screen images show versions 0.11.1 to 0.11.4.

Revision	Change
A, November 2026	First issue, for software version 0.11.1.
B, November 2026	For software version 0.11.3. The displacement diagram marks each problem with a warning sign over the curve and a dashed line through the plots, red for an error and amber for a warning (version 0.11.2). Each picture has a caption. The firmware uses the coordinated motion controller of the EnhancedClearCoreLibrary (version 0.11.3).
C, November 2026	For software version 0.11.4. The board's EtherNet/IP identity uses vendor ID 65000 (the PLC chapter's identity table).

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Preface

Purpose of This Manual

This manual tells you how to use the Electronic CAM Workbench. With the Workbench you design the electronic cam profiles of a machine, check them, and run them on a Teknic ClearCore that moves Teknic ClearPath servos. The manual also tells you how to set up the ClearCore, its inputs and outputs, homing, and limits, and how to control the machine from a PLC.

The mathematics of the profiles and the operation of the firmware are in the companion document, *Cam Law Mathematics and ClearCore Theory of Operation*, ECW-TOP-100.

Audience

This manual is for the engineers and technicians who design, commission, and operate a machine with the Workbench. You must know the mechanics of your machine and the basics of servo motors. You do not need to know the mathematics of cams. The glossary at the end of the manual defines the terms that the manual uses.

How to Use This Manual

Table P.1 shows where to start for each task.

Table P.1. Where to start

Task	Read
Make a first project with the guided setup	Chapter 1
Learn the parts of the main window	Chapter 2
Select a crank or a timeline master	Chapter 3
Draw and edit a profile, and read its analysis	Chapters 4 and 5
Set the motor, mechanics, homing, and limits of an axis	Chapter 6
Switch outputs during the cycle	Chapter 7
Set up a flying saw with a variable piece length	Chapter 8
Connect to the ClearCore, upload, and run	Chapter 9
Learn from an example machine	Chapter 10
Save, export, or print	Chapter 11
Look at the motor signals with the scope	Chapter 12
Find a shortcut key	Chapter 13
Find the cause of a problem	Chapter 14
Control the machine from a PLC	Chapter 15

Conventions

Safety Notices

△ WARNING

A warning tells you about a condition that can cause injury or death. It also tells you about a condition that can make machinery move when you do not expect it.

CAUTION

A caution tells you about a condition that can cause damage to equipment or incorrect motion.

NOTE: A note gives information that helps you. It does not tell you about a hazard.

△ WARNING

The Workbench and its firmware are not certified for safety functions. Install a physical emergency stop that removes the power to the motors before you run a machine.

Typography

Table P.2. Typographic conventions

Example	Meaning
Upload, **File › Save**	A button, a field, or a menu command in the Workbench.
Ctrl + S	Keys to push together.
`.ecam`, `axjog`	A file name, a file type, or a command of the ClearCore console.
no estop	Text that the Workbench shows.
Building a profile (chapter 4)	A reference to a different part of this manual. In the PDF, the text is a link.

Related Documents

- *Cam Law Mathematics and ClearCore Theory of Operation*, ECW-TOP-100: the mathematics of the motion laws and the operation of the ECam firmware.
- *QuickEIP Programming and Implementation Manual*, QE-PIM-200: the EtherNet/IP stack of the firmware.
- `README.md`: an overview of the Workbench, and how to build it.
- `Firmware/PROTOCOL.md`: the console commands of the ECam firmware and the EtherNet/IP assemblies.

CHAPTER 1

Getting Started

The **Electronic CAM Workbench** designs, checks and runs electronic cam profiles on a Teknic **ClearCore**. A follower motor moves as a function of a **crank** (an encoder on the machine, or a crank motor the ClearCore drives itself) or of a **timeline** (time from a trigger). You draw the motion, the app checks it (velocity jumps, motor speed, limits, cam pressure angle), and the ClearCore keeps it in flash and runs it from its 5 kHz sample interrupt, with nothing streamed while it runs.

CAUTION

Experimental. Not certified for production or safety-critical use. Keep a physical emergency stop in the circuit (the firmware can watch an input as one: A-9 unless you pick another).

1.1 The Start Page

After the splash screen the **start page** offers the ways in:

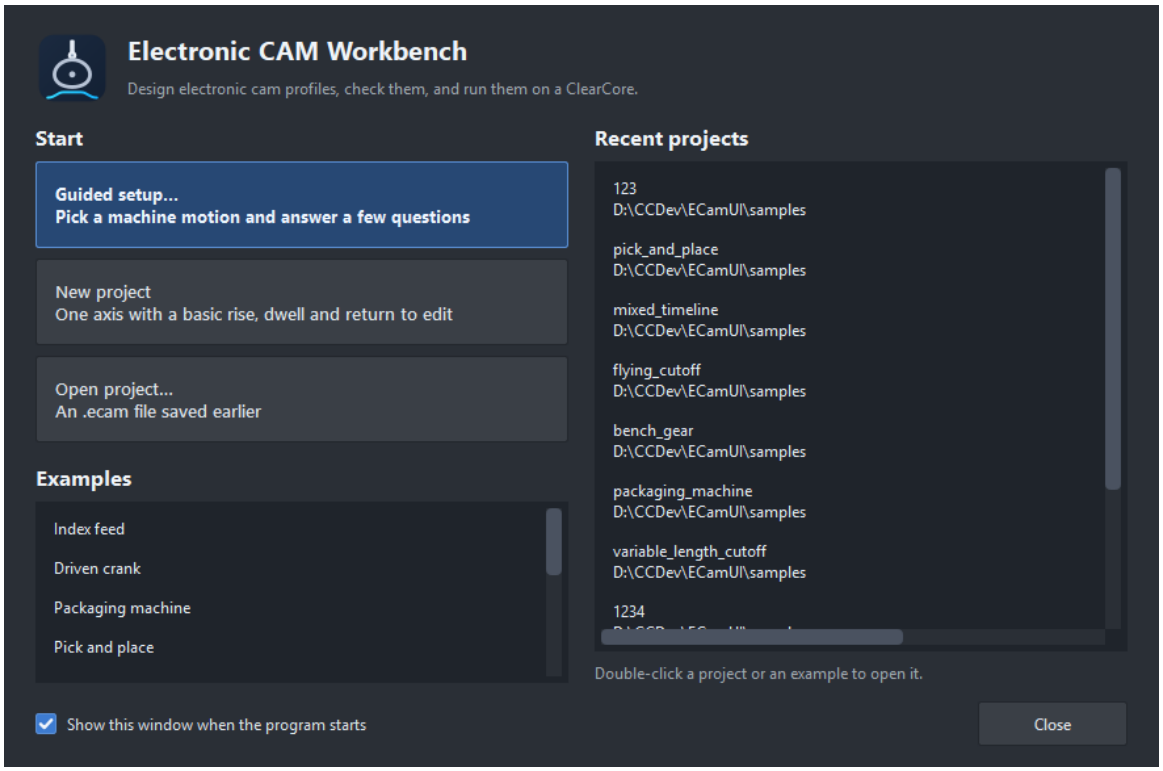


Figure 1.1. The start page: guided setup, a new project, an existing file, a bundled example or a recent project.

- **Guided setup...** walks you from a machine motion to a working project (below). The best

start.

- **New project**: one axis with a basic rise, dwell and return, ready to edit by hand.
- **Open project...**: an `.ecam` file saved earlier.
- **Examples**: projects built into the program (double-click one). Save it to keep your own copy.
- **Recent projects**: the last ten you opened or saved.

Untick *Show this window when the program starts* to go straight to a new project; **File › Start page...** brings it back.

1.2 Guided Setup

Guided setup (start page, or **File › New from guided setup...**, Ctrl + Shift + N) asks a few plain questions, one step at a time, and builds the project. Each step explains what it needs.



Figure 1.2. Step 1: pick the motion closest to your machine. The preview shows its shape.

Template	Master	For
Rise, dwell, return	crank	a pusher, an ejector, a classic cam follower
Index feed	crank	a conveyor or roll feed that advances a fixed length each cycle
Oscillate	crank	continuous out and back: a traverse, a wrapping arm

Template	Master	For
Triggered move	timeline	out, hold and back once per input: pick and place, a gate
Flying cut-off	encoder on the line	a carriage that rides the material while a saw cuts

Then the steps ask what times the cycle (an encoder, a crank motor, a trigger, or the line), the motor and its mechanics (one step per motor), and the motion itself, with a live preview and the peak velocity, acceleration and motor speed:

The **Review** step lists the settings and any problems (two things on one motor connector, a velocity jump, a limit exceeded, a fast motor) and what to do next. **Create project** opens it in the editor, unsaved.

1.3 From Idea to Motion in Five Steps

1. **Make the profile**: guided setup, an example, or draw it (Building a profile (chapter 4)).
2. **Check it**: the Analysis page and the warning signs in the diagram flag jumps and limits (Analysis (section 4.5)).
3. **Save it**: **File** › **Save** (Ctrl + S).
4. **Upload it**: in the board strip at the bottom, **Connect** and **Upload** (Running on the ClearCore (chapter 9)).
5. **Run it safely**: tick **Simulate** and **Start** first (nothing moves), then untick Simulate, **Enable** and **Start**.

> **NOTE:** New to electronic cams? Read The window (chapter 2) and Masters (chapter 3), then try a worked example (chapter 10).

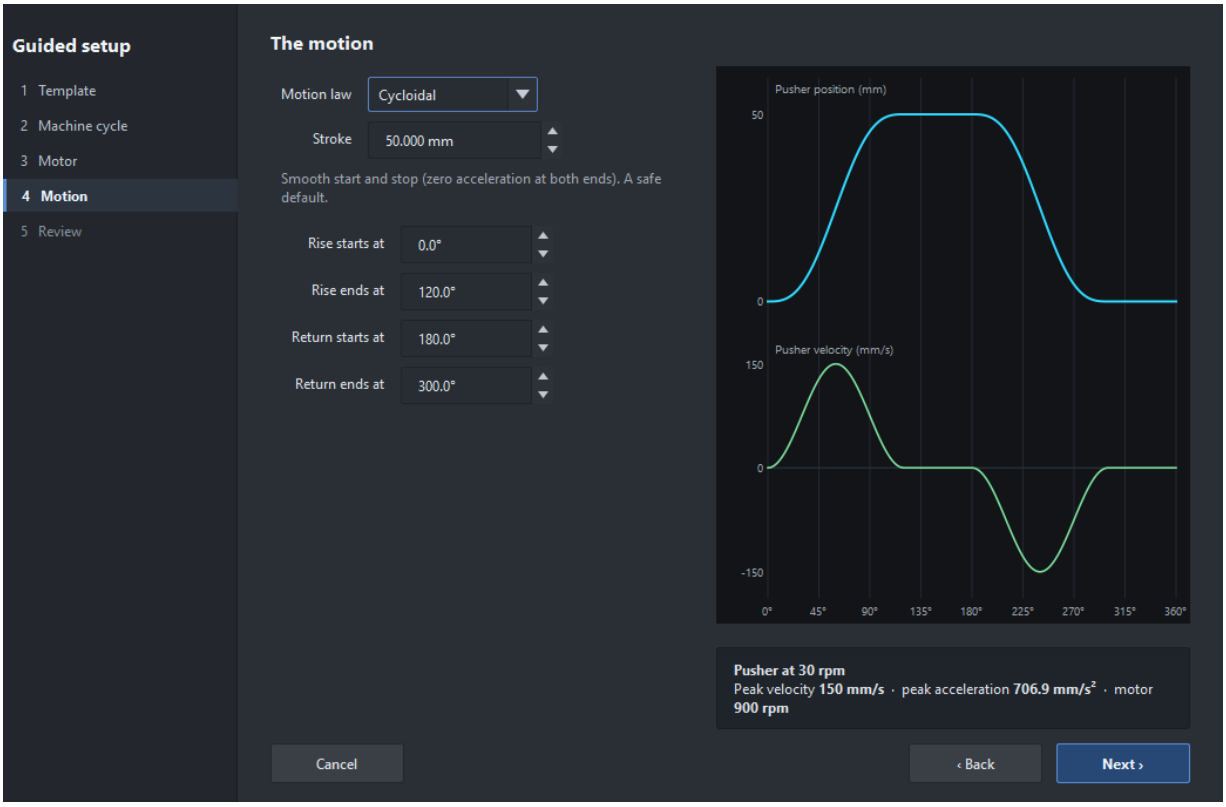


Figure 1.3. The motion step: stroke, where in the cycle it moves, and the motion law. The card below the preview shows the peaks.

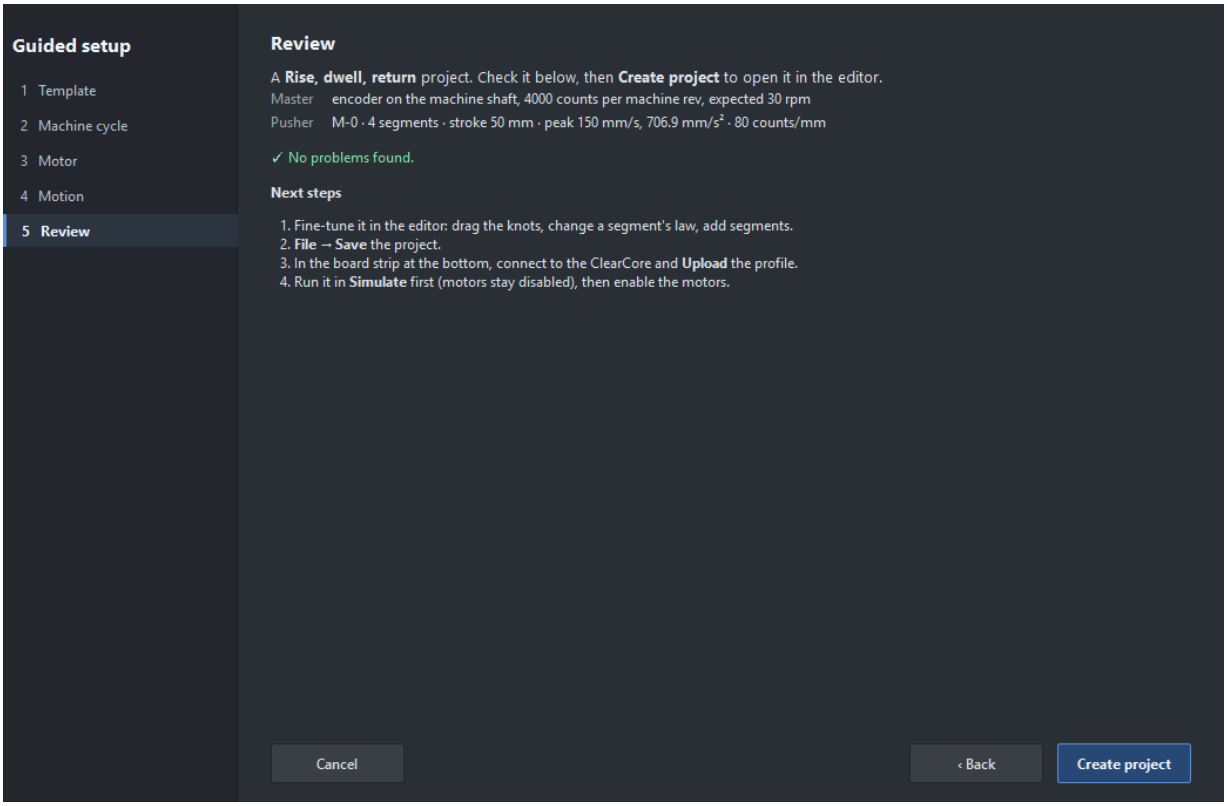


Figure 1.4. The Review step: the settings, any problems, and what to do next.

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Chapter 2

The Window

Everything about one project is on one screen, and the layout is remembered. It fits a 1920 × 1080 screen with nothing to scroll.

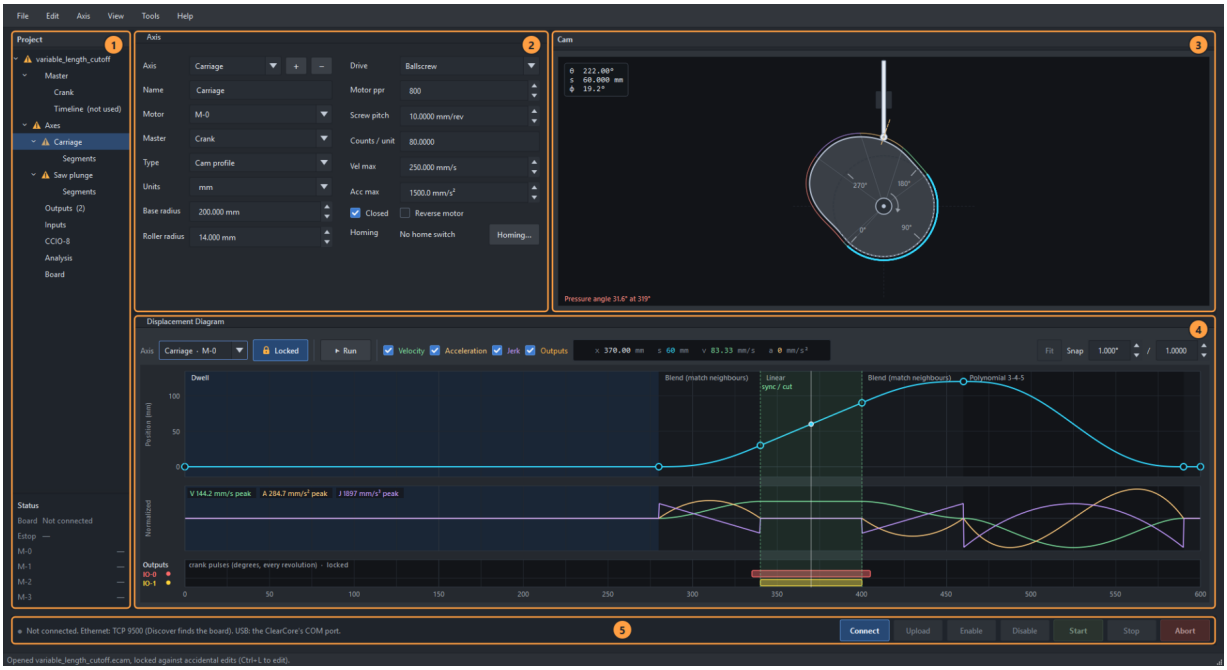


Figure 2.1. A flying saw (cut-length mode) open in the editor, the carriage axis selected.

#	Part	What it is for
1	**Project tree**	The project's master, axes and their segments, outputs, inputs, CCIO-8, analysis and board. Click a node to open it in the detail pane. Under it, the Status (section 2.2) panel. F9 or « hides both.
2	**Detail pane**	Whatever the tree has selected: *Crank*, *Timeline*, an axis, its *Segments*, *Outputs*, *Analysis* or *Board*. The diagram stays in view while you edit.
3	**Cam / Timing**	The selected crank axis drawn as a plate cam, or the Timing view of every timeline axis, whichever the selected axis is on (Ctrl + T: the Timing view for a crank axis too, in a project with both). Shift + F9 (**View › Cam / Timing view**) or » folds it away and gives the detail pane its width.
4	**Displacement Diagram**	The selected axis over its master, the full width of the window: position, then velocity, acceleration and jerk, then the Outputs band. Drag knots here.
5	**Board strip**	Whether the board is there, what it is doing, and the buttons that run it, always in view (see The board strip (section 9.2)).

Drag the line between the top row and the diagram to share the height differently. On a short window the top row gives way and its pages scroll, and it comes back when there is room.

2.1 The Project Tree

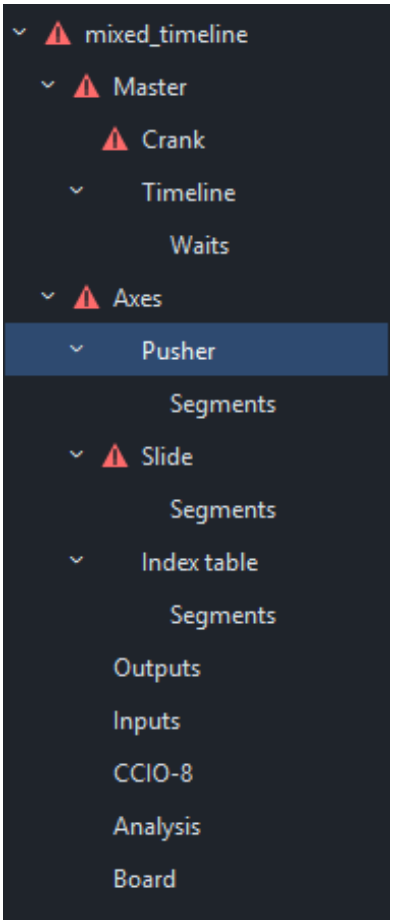


Figure 2.2. The project tree. A triangle marks a node with a problem.

The tree is the project at a glance, with CCIO-8 (section 9.10) inputs under their own node. Clicking an axis selects it everywhere, and the tree follows the diagram's **Axis** list; clicking a segment in the diagram opens *Segments*. Right-click an axis for the Axis menu.

A **triangle** marks what the project review found: red for an error, amber for a warning. Hover over the node to read it. A problem with an axis is on that axis, a problem with the master on Crank or Timeline, and one that belongs to the whole project, such as two axes on one motor, on the project. Headings and the project show the worst thing under them, so folding a branch can't hide a red one. *Analysis* has the details for the selected axis.

2.2 The Status Panel

Under the project tree, the board as it is now: what it is doing, the estop (**OK**, **Open**, or **Off** when the board ignores it, see Estop (section 9.5)), and each motor connector, M-0 to M-3. A motor reads **Disabled**, **Enabled** (with its HLFB up), **Enabled, no HLFB** (enabled but the motor isn't saying it is ready: still enabling, or not connected), or **Fault**. When the open project is the one on the board, each motor shows its axis's name, or *Crank*. A latched fault shows under the motors. Drag the line above it to give the tree more or less room. With no board connected it shows dashes.

2.3 The Displacement Diagram



Figure 2.3. Position (top), velocity and acceleration (middle, normalized) and the Outputs band (bottom). The green band is the saw's sync window.

The bar above the diagram holds, from the left:

- **Axis**: which axis the diagram (and the Segments table) shows. Ctrl + PgUp / Ctrl + PgDn step through them.
- **Lock** (Ctrl + L): locks the profile against accidental edits (see Locking (section 4.4)).
- **Run** (Space): turns the crank at the master speed, or plays the timeline, in the app only.
- **Velocity**, **Acceleration**, **Jerk**, **Outputs**: what the diagram shows.
- The **readout**: at the cursor, the master position (θ in degrees, t in seconds, or x in mm of line in cut-length mode), then position s, velocity v and acceleration a per second. While you drag a knot it shows the knot.
- **Fit**: back to the whole cycle after zooming (it shows the zoom).
- **Snap**: the grid knots snap to while dragging, in the master's units and in position.

On a narrow window the readout, Fit and Snap wrap to a second row.

2.3.1 Zoom and Pan

Ctrl +wheel zooms the master axis about the mouse (all bands together), Ctrl + / − about the cursor, Ctrl + 0 or **Fit** shows the whole cycle. Middle-drag or Shift +wheel pans while zoomed.

2.4 The Cam View

A crank axis drawn as the mechanical plate cam it replaces: the cam turns clockwise, the follower is a translating roller at 12 o'clock, coloured by segment. The dashed orange line is the contact



Figure 2.4. Zoomed ×3 on the saw's sync: the grid gets finer, the knots and pulses stay editable.



Figure 2.5. The cam view: the crank axis drawn as the plate cam it replaces, with the follower at 12 o'clock.

normal (the pressure angle). Drag the cam to turn it, the wheel steps 1° (Shift: 10°). The box in the corner reads the crank angle θ, position *s* and pressure angle φ. Base and roller radius (Axis page) only change this drawing and the DXF export.

2.5 The Timing View

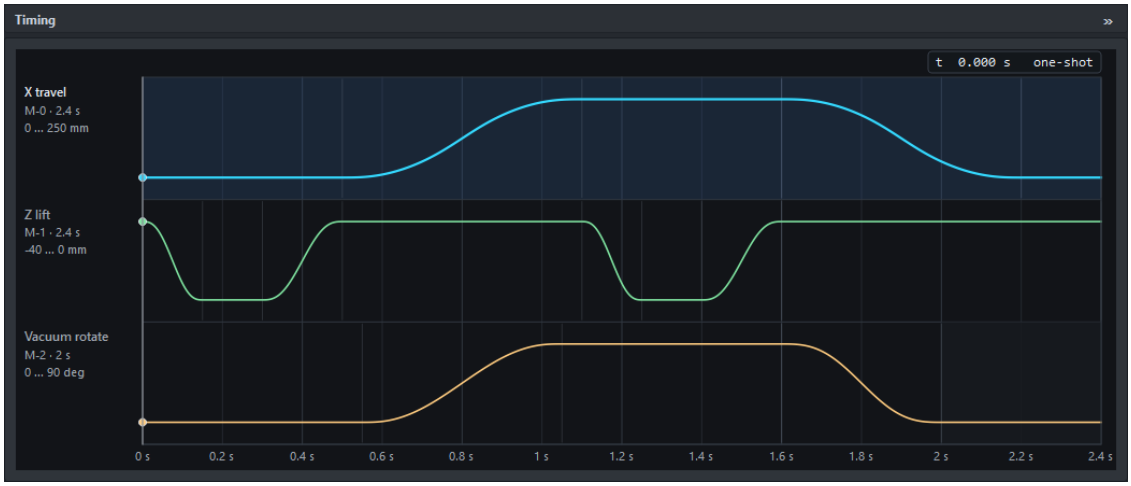


Figure 2.6. The Timing view: every timeline axis on one time axis, with the output pulses.

Every timeline axis on the shared timeline, with the timeline's output pulses. It shows when a timeline axis is selected. In a project with crank and timeline axes, **View › Timing view for crank axes** (Ctrl + T) keeps it up while a crank axis is selected too. Click an axis's row to select it; scrub to move the cursor.

2.6 Menus

Menu	Holds
File	New, guided setup, open, recent, examples, start page, save, exports (CSV, C header, cam DXF, Excel workbook).
Edit	Undo / redo (every edit; a drag is one step), split and delete segment.
Axis	Add, duplicate and remove axes (up to four, one per motor connector).
View	Project tree (F9), Cam / Timing view (Shift + F9: shows or hides the pane), Timing view for crank axes (Ctrl + T, projects with both kinds of axis).
Tools	ClearPath scope (Ctrl + Shift + O).
Help	This guide (F1), worked examples, keys, the scope, About.

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CHAPTER 3

Masters: Crank and Timeline

Every axis follows a **master**. One firmware runs up to four axes (one per motor connector, M-0 to M-3), and each axis picks its master: crank axes and timeline axes can run together.

Master	x is	Repeats	Typical use
Crank	the crank angle, 0 to 360°	every revolution	anything tied to a machine shaft or a line
Timeline	seconds from a trigger	once per trigger, or repeating	moves started by an input: pick and place, gates

3.1 The Crank

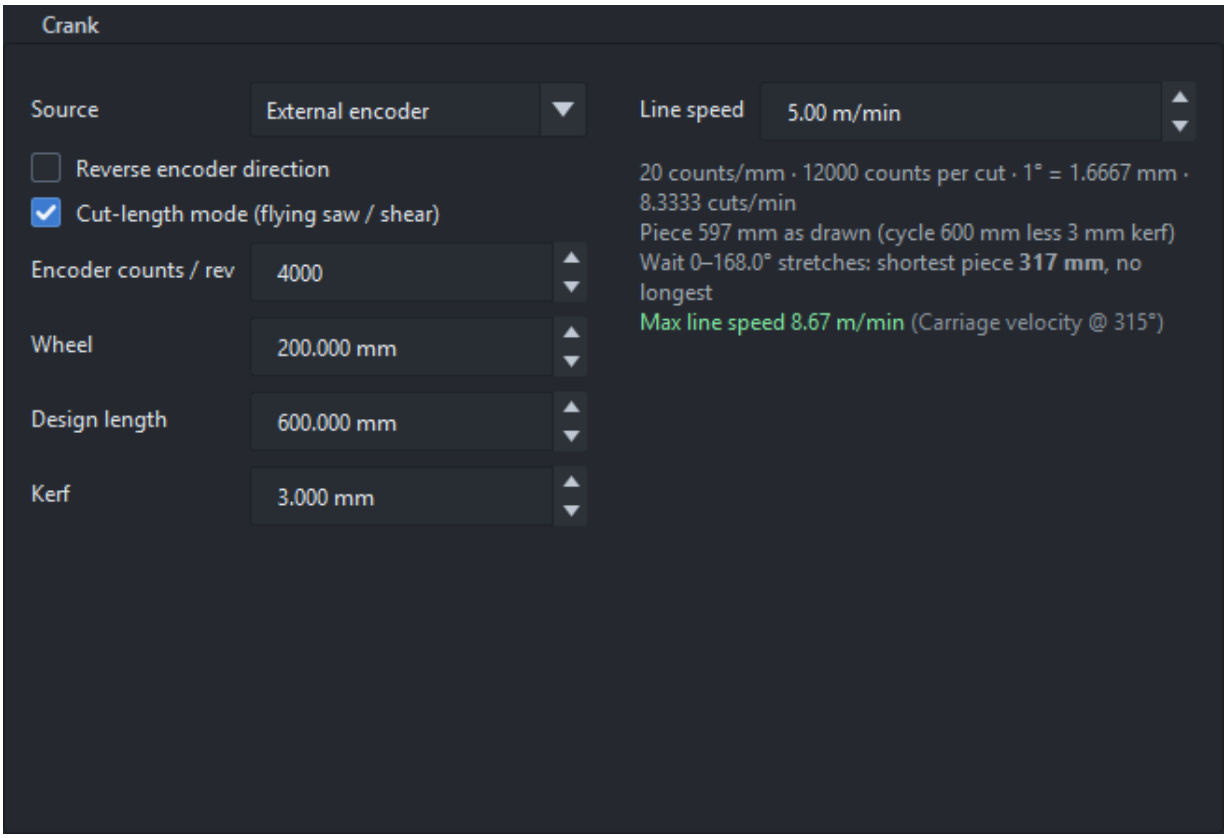


Figure 3.1. The Crank page, here in cut-length mode with an encoder on the line.

The crank is either:

- an **external encoder** on the Encoder Adapter: the machine runs on its own and the followers track it. **Encoder counts / rev** is **4 × the encoder's line count** (every quadrature edge

counts). **Expected speed** is only used for the analysis and the preview; or

- a **driven crank**: a motor the ClearCore runs itself at the **Run speed**, ramping at the **Ramp** (rpm/s) up to the motor's **max**.

The **gear ratio** is motor (or encoder) turns : crank turns as a fraction, so a gearbox that is not a whole ratio stays exact (50 : 7). With axis velocity / acceleration limits set, the page gives the **highest safe crank speed**, including the extra acceleration followers feel while a driven crank ramps.

3.1.1 Bringing an Encoder Up

1. Connect and watch the **Encoder** row on the Board page: it should count **up** as the machine moves forward. If it counts down, tick **Reverse encoder direction** (Crank page) and Upload again; do not swap wires.
2. Press **Zero** and turn the encoder one revolution: the row reads 1.000 rev.
3. Upload, tick **Simulate**, **Start**, and move the machine: the cam view and the diagram follow.

> **NOTE:** No encoder yet? Tick **Simulate**, **Start**, then **Run** in the **Sim encoder** row of the Board page row: the board moves a simulated count and everything follows it as it would a real one.

3.2 The Timeline

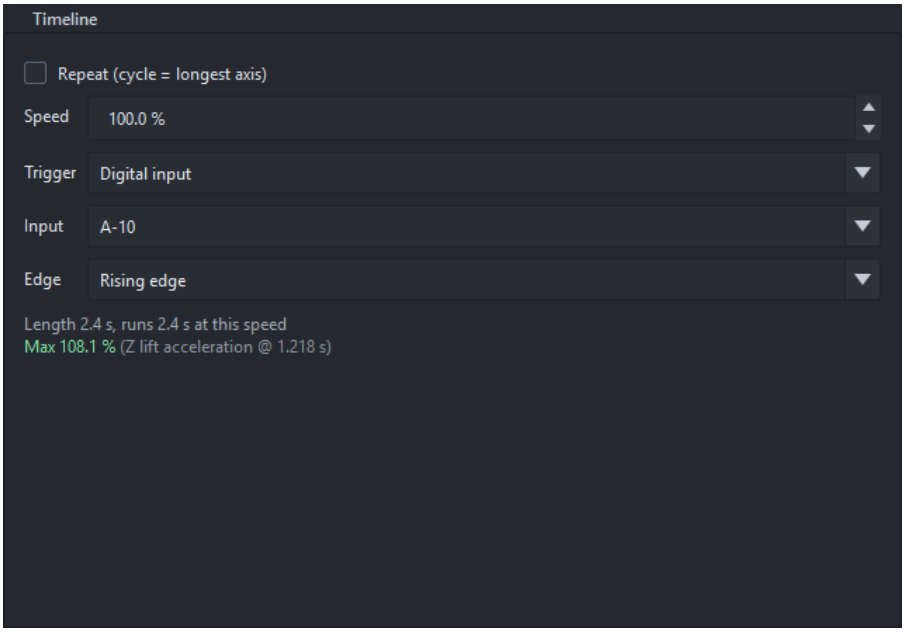


Figure 3.2. The Timeline page: repeat, speed override and the trigger.

On the timeline x is time in seconds from a $t = 0$ shared by every timeline axis. Each timeline axis has its own **duration** (drag its last knot sideways, or set Duration in the Axis page) and rests at its end position after it.

- **Trigger**: a digital input and edge, or a command (the **Trigger** button in the board strip). The input list greys out inputs that cannot be used (the Encoder Adapter's DI-6..DI-8 while an encoder is the master, an IO pin used as an output, the board's estop input).
- **One-shot** or **repeat**: one-shot stops at the end (*done*; Trigger runs it again), repeat wraps to 0.
- **Speed override**: the timeline's speed, 100 % = as drawn. With limits set, the page gives the highest safe override.

The timeline's length is the longest timeline axis or the last timeline output pulse, whichever ends later.

3.3 Waits

A **wait** holds the timeline at a point until an input says go. It is how a timeline machine checks that something happened before it carries on: the part is gripped before the lift rises, the clamp is closed before the saw comes down, the PLC has cleared the place position before the arm swings over it.

At a wait, every timeline axis and every timeline output stays exactly where it is. When the input arrives the timeline runs on from there, as drawn, as if it had never stopped: each wait only adds the time it held to the cycle. Crank axes, in a project that has both, carry on: a wait holds only the timeline.

3.3.1 An Example: Pick and Place

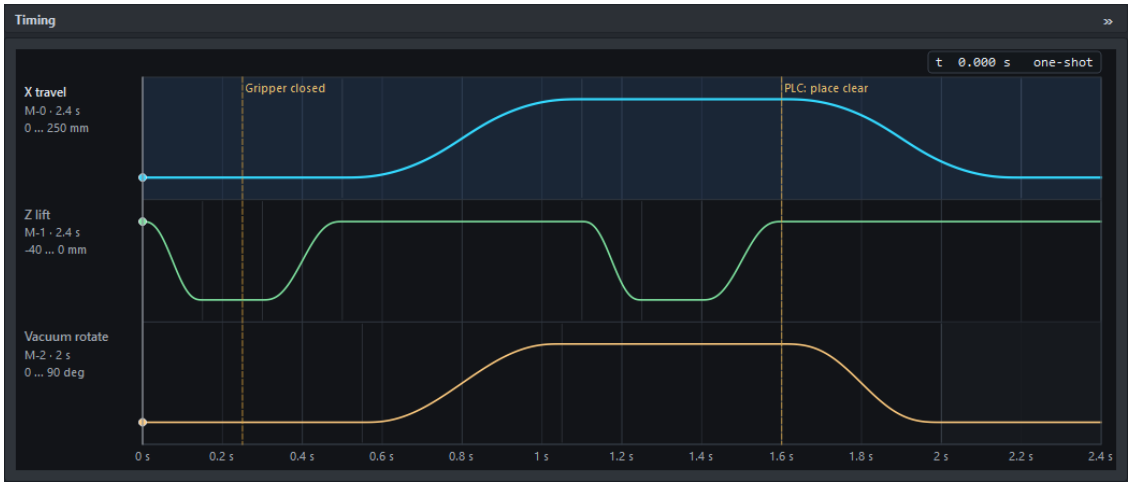


Figure 3.3. The pick and place with two waits, drawn as dashed lines across the Timing view.

In the *Pick and place* example the Z lift goes down, the vacuum comes on (IO-0, from 0.3 s), and the lift rises with the part. Nothing on the timeline knows whether the part was really picked. Two waits fix that:

1. **Gripper closed** at 0.25 s, on IO-2 (the vacuum switch), *On*, with a 2 s timeout. The lift is down and still; the timeline holds there until the vacuum switch says the part is held. If it never does, after 2 s the machine stops with a fault instead of swinging an empty gripper across.

2. **PLC: place clear** at 1.6 s, on PLC bit 0, no timeout. Before the arm comes back, the PLC confirms the place position is clear; the machine waits as long as it takes.

To build it: open **Timeline › Waits** in the project tree, move the cursor near where the wait should be, and press **Add**. Or right-click the Timing view there: *Wait for an input here…*. Then fill in the row.

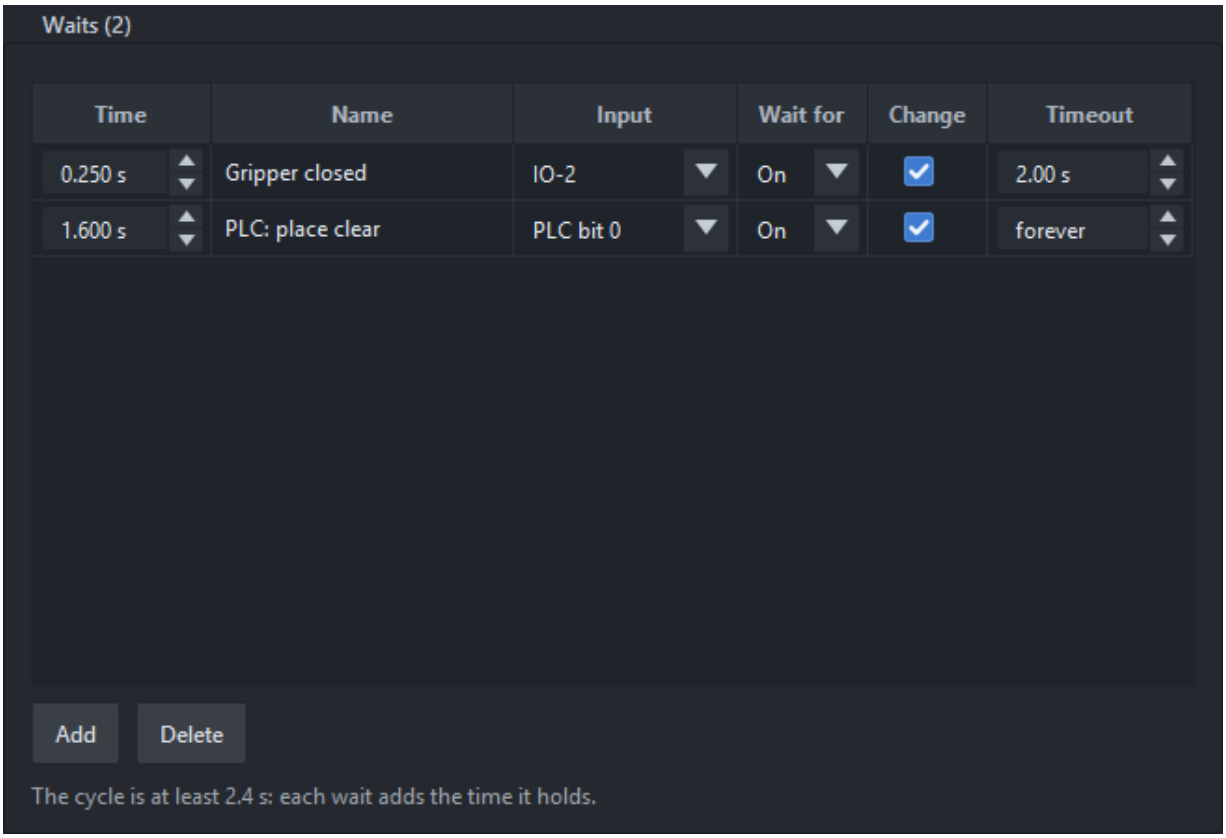


Figure 3.4. The Waits page of the pick and place: one wait on a board input, one on a PLC bit.

3.3.2 The Settings

Time	Seconds into the cycle, as drawn. A wait goes only where every timeline axis is at rest (the board stops the clock at once, which is gentle only there): **Add** and the right-click put it at the nearest such time, and Analysis marks one where an axis moves as an error. To move a wait, drag its line in the Timing view: it turns red where an axis is moving and, dropped there, snaps to the nearest time they all rest.
Name	What the operator sees while it waits: *Waiting for Gripper closed*.
Input	A board input (IO-0 … A-12), a CCIO-8 input, or a **PLC bit** (0 … 31) that the PLC sends in its output assembly (the PLC chapter (chapter 15)). An input that already does something else (an output, the trigger, the estop, a home or limit, an operator input) can't be one. A wait's input shows *Timeline wait* on the Inputs page.
Wait for	*On* (the input closed / the bit set) or *Off*.

Change	Ticked (the default): the wait passes only if the input has been at the *other* level since this wait last passed (since Start, the first time), and is at the wanted level now. So a sensor stuck *On* can't let the next cycle through: the wait holds, and times out if it has a timeout. The change can come any time before the timeline gets there: a gripper that closes early isn't missed. Unticked: the level alone, for a steady condition that is simply true or not, such as a guard door closed or the PLC's go.
Timeout	How long it may hold, or *forever*. Past it, the machine stops at the wait with the fault *wait 1 (IO-2 on) timed out at 0.250 s*, the motors still enabled and holding.

3.3.3 How It Runs

- **Exactly at its time.** The timeline's clock stops on the wait's time, not near it. Everything drawn before that time has happened; nothing at it has, so an output pulse that starts at the wait's time starts when the wait passes ("wait, then do").
- **Every cycle.** Each wait holds once a cycle, one-shot or repeating.
- **Fast.** A wired input is seen within a millisecond (it is filtered over 0.6 ms against bounce). A PLC bit arrives with the PLC's next packet and scan: 20 ms or more. A PLC-bit wait never passes while the PLC isn't connected and in Run, whichever level it wants.
- **Stop** during a wait stops the machine there. It does not finish the cycle: that would make the very moves the wait was holding back. Start then waits until the timeline axes are back at their start (**To cam position** in Axis jog), as after any cut-short cycle. **Abort**, **Disable**, the estop and a lost PLC connection work as always.
- **Hand jog** ignores waits. **Simulate** holds on the real inputs, and **Skip wait** on the Board page (Timeline row) passes the one holding, so a timeline can be tried without its sensors.

3.3.4 What You See

While a wait holds, the board strip's timeline line says *Waiting for Gripper closed (IO-2 on), 3.2 s*, and the Timing view lights that wait's line. The PLC sees TimelineState 6 (waiting), StatusFlags bit 14, which wait (WaitIndex) and for how long (WaitedMs): with no timeout, the PLC can raise its own alarm after however long suits the machine.

> **NOTE: A wait that never passes:** look at its input on the Inputs page (State) while it waits. With **Change** ticked, the input must also go to the other level once: a sensor stuck on holds it, which is what the setting is for. A PLC-bit wait needs the PLC connected and in Run.

3.4 Changing an Axis's Master

Changing an axis's master keeps its motion: crank → timeline maps one revolution at the current crank speed onto seconds; timeline → crank stretches the duration to 360°.

For a flying saw, laser or shear whose cut length changes on the fly, see cut-length mode (chapter 8).

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Chapter 4

Building a Profile

An axis's motion is a list of **segments** over its master (0–360° on the crank, 0 to the duration on the timeline). Each segment has an **end** (where it stops on the master), an **end position** (in the axis's units) and a **motion law**; it starts where the previous one ended.

4.1 Knots

The boundaries between segments are **knots** (the circles in the diagram). Knot 0 is at $x = 0$ (the start position); the last knot is at 360° (crank) or at the duration (timeline).

- **Drag** a knot to move it: sideways changes when, up and down changes where. Shift drags vertically only, Ctrl horizontally only, Alt without snapping.
- A **dwell** (no motion) moves with its knots: its end is tied to its start.
- **Double-click** in a segment to **split** it there; **right-click** for its law or to delete it.
- Click or drag on empty space to **scrub** the master (the cursor line).

> **NOTE:** Every edit can be undone (Ctrl + Z); a whole drag is one step.

4.2 Closed and Open Profiles

- **Closed** profiles return to the start every revolution (the last knot is tied to knot 0): pushers, cams, oscillators.
- **Open** profiles advance by *end* − *start* every cycle: index feeds and conveyors.
- A timeline axis must start and end at rest.

4.3 The Segments Table

One row per segment. Pick the law in the first column; double-click a cell to type an exact end, span, position or rise. **Split** and **Delete** work on the selected row.

4.4 Locking

The **lock** in the diagram's bar (or **Edit › Lock profile**, Ctrl + L) guards the profile against accidental edits. While it is locked, knots and pulses can't be dragged, double-click and right-click don't split, delete or change laws, and the Segments and Outputs tables are read-only. Scrubbing, zoom, pan, undo / redo and **Run** still work, and the pages in the detail pane are not locked.

- A project **opened** from a file, a bundled example or a **download** from the board starts **locked**: look first, unlock to change it.

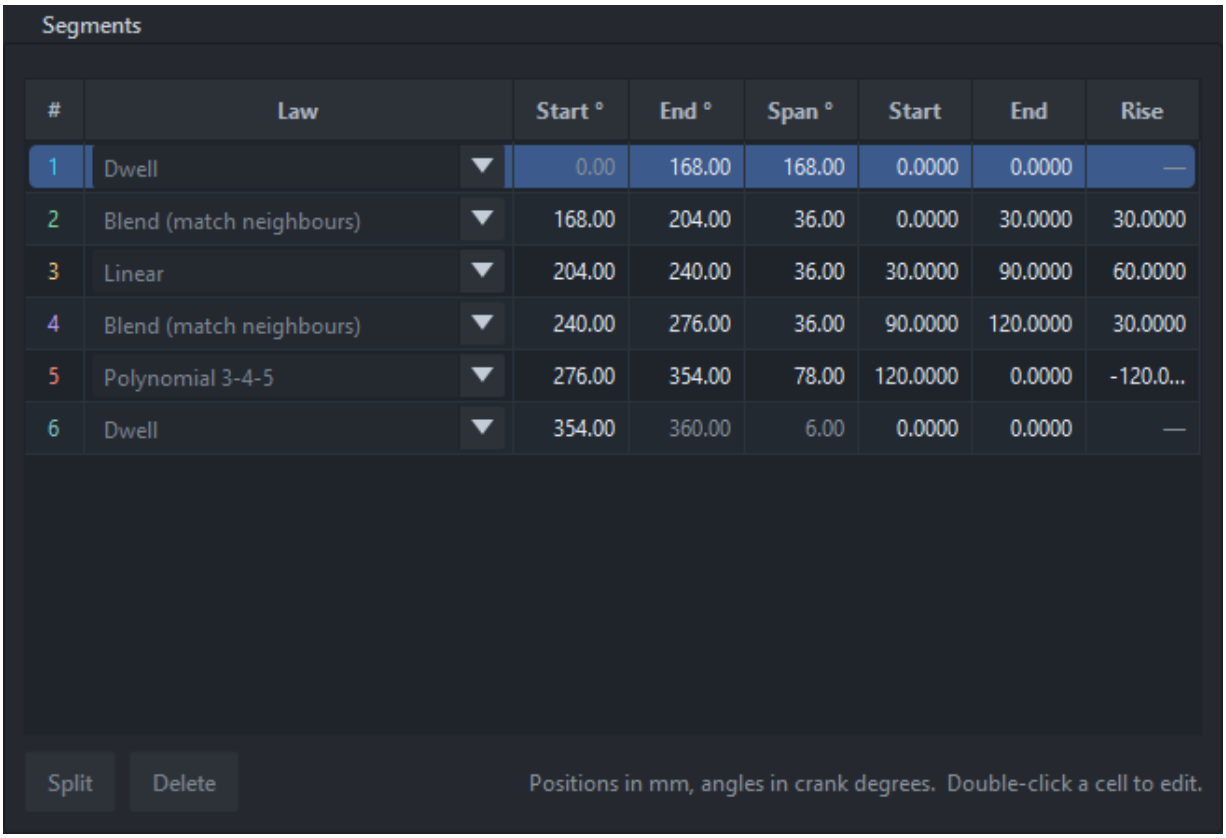


Figure 4.1. The Segments table of the flying saw's carriage: one row per segment.

- A **new** project or one from the guided setup starts **unlocked**.

The lock is not saved in the file.

4.5 Analysis

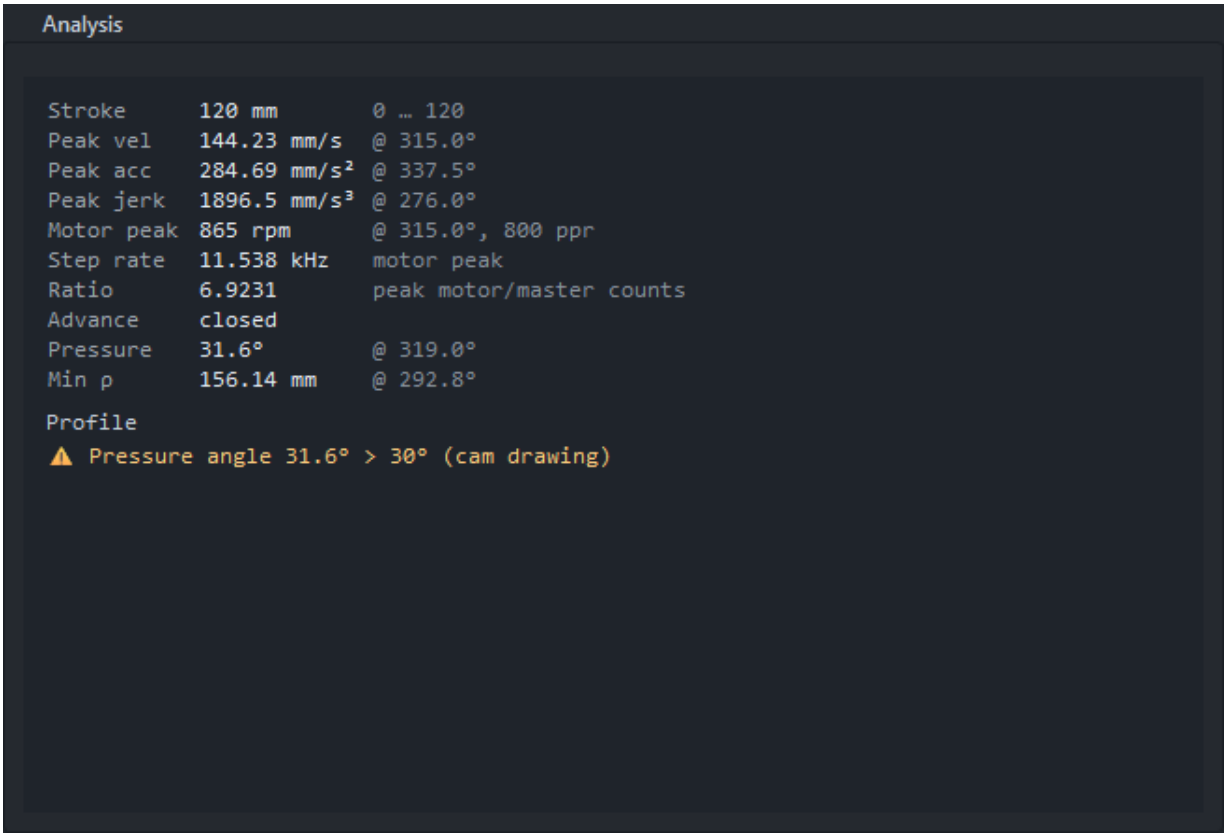


Figure 4.2. The Analysis page: the peaks, motor speed and warnings of the selected axis.

For the selected axis, at the master speed:

- **Stroke**, **peak velocity**, **acceleration** and **jerk** (with where they happen);
- **Motor peak** (rpm) and **step rate** (kHz): check them against the motor's rating;
- on the crank, the motor/master counts **ratio**, the **advance** per revolution, the cam's largest **pressure angle** and smallest **curvature radius**; on the timeline, the duration and the highest safe **speed override**;
- **Warnings**: velocity steps (infinite acceleration: a hard knock), acceleration steps (infinite jerk), pressure angle above 30° and undercut, motor connectors used twice, speeds above the limits.

The diagram marks each problem where it happens: a **warning sign** over the curve and a dashed line down through the plots, red for an error (a velocity step, a velocity or acceleration over the axis's limit) and amber for a warning (an acceleration step, a spline overshoot, a pressure angle above 30°, undercut). Hover over the sign for what it is and what to do.



Figure 4.3. A pusher with a Linear rise and a Harmonic return: velocity steps (red) where the rise starts and stops, acceleration steps (amber) at the ends of the return, and the return's velocity over the axis's limit (red).

4.6 Several Axes

Axis › Import points as spline replaces the axis's segments with a Cubic spline (chapter 5) through the points in a file: a CSV or text file with the master value in the first column and the position in the second (a table from File › Export reads back too; lines starting with #, and a heading line, are skipped). The values in the first column map onto the axis: first to last is the whole revolution on the crank, and the duration on the timeline. Up to 65 points become knots; with more, the Workbench picks the 65 that keep the curve closest, and says how far it ends up from your points. If the last position equals the first the axis is closed, otherwise open. One undo puts the axis back. Try it with `samples/spline_points.csv`.

Axis › Add axis (or Duplicate) adds a follower; each needs its own motor connector. Pick the axis in the diagram's **Axis** list (or the Axis page). Every crank axis shares the crank, every timeline axis the timeline.

Which law to use? See Motion laws (chapter 5).

Chapter 5

Motion Laws

A segment's **law** is the shape of its move from one position to the next. The choice decides how hard the motor and the machine work: a law that starts and ends gently lets the machine run faster without knocks.

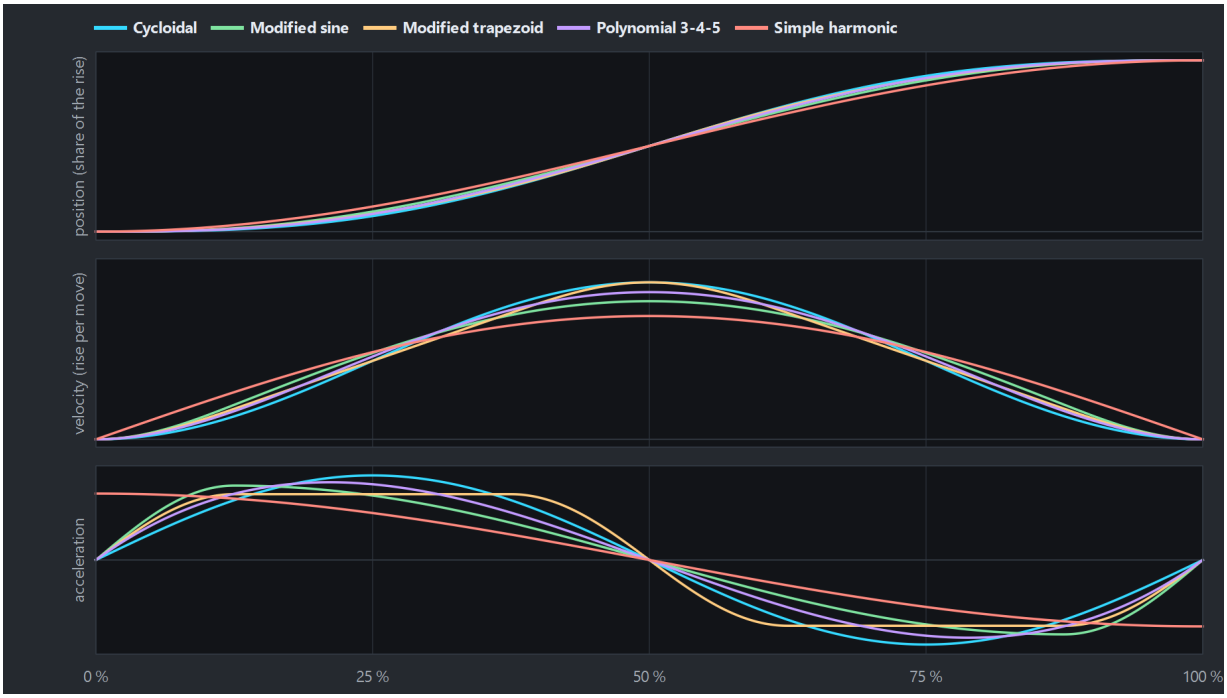


Figure 5.1. Five laws over one move, at the same rise and time: position, velocity and acceleration.

5.1 The Laws

Law	Ends at rest	Use it for
Dwell	—	no motion; its end is tied to its start
Cycloidal	yes, zero acceleration	a safe default: smooth start and stop
Modified sine	yes	heavier loads and index drives: lower peak velocity
Modified trapezoid	yes	fast, light moves: the lowest peak acceleration
Polynomial 3-4-5	yes	a smooth polynomial, a little gentler on velocity than cycloidal
Polynomial 4-5-6-7	yes, and zero jerk	the smoothest, for slightly higher peaks
Simple harmonic	yes, but acceleration jumps	oscillating out and back
Linear	no	constant speed: riding a conveyor or the line between other moves

Law	Ends at rest	Use it for
Blend (match neighbours)	no	joining a dwell to a linear sync smoothly (a flying saw's ramp)
Cubic spline	no	a smooth curve through points you give (measured data, a drawn shape)

5.2 Choosing

- **Start with Cycloidal.** It starts and ends with zero acceleration, so it joins dwells without a knock.
- **Peaks too high?** Modified trapezoid lowers the peak acceleration (motor torque); modified sine lowers the peak velocity (motor speed). Or give the move more of the cycle.
- **Linear** has velocity jumps at its ends unless its neighbours match its speed. Put a **Blend** segment on each side: a Blend starts and ends at its neighbours' velocities with zero acceleration, so dwell → Blend → Linear → Blend → return runs smoothly.
- Between two rests a Blend is the 3-4-5 polynomial.
- **Cubic spline** is the one law that works across segments: consecutive Spline segments make a single smooth curve through all their knots, so moving one knot reshapes its neighbours. Use it when the shape comes from points rather than from a rise and a dwell.
  - Next to any other segment it takes that segment's velocity, so there is never a velocity step. It can still have an *acceleration* step there, and the Analysis page says so. A Blend at the end of the run removes it.
  - Make every segment of a repeating axis a Spline and the curve closes on itself with no step at all, through the wrap too.
  - A spline can swing past the points it goes through, most of all after a steep rise into a flat stretch. The Analysis page warns when it passes your highest or lowest knot by over 2% of their range: add a knot, or end the run with a dwell.

> **NOTE:** The Analysis page flags any velocity step (infinite acceleration) and acceleration step (infinite jerk) the laws you pick leave at a knot; the diagram marks them with warning signs.

The formulas are in `docs/MATH.md` in the project.

Chapter 6

Axes and Mechanics

The **Axis** page holds the selected axis's name, master, motor connector, units, mechanics, limits and cam drawing.

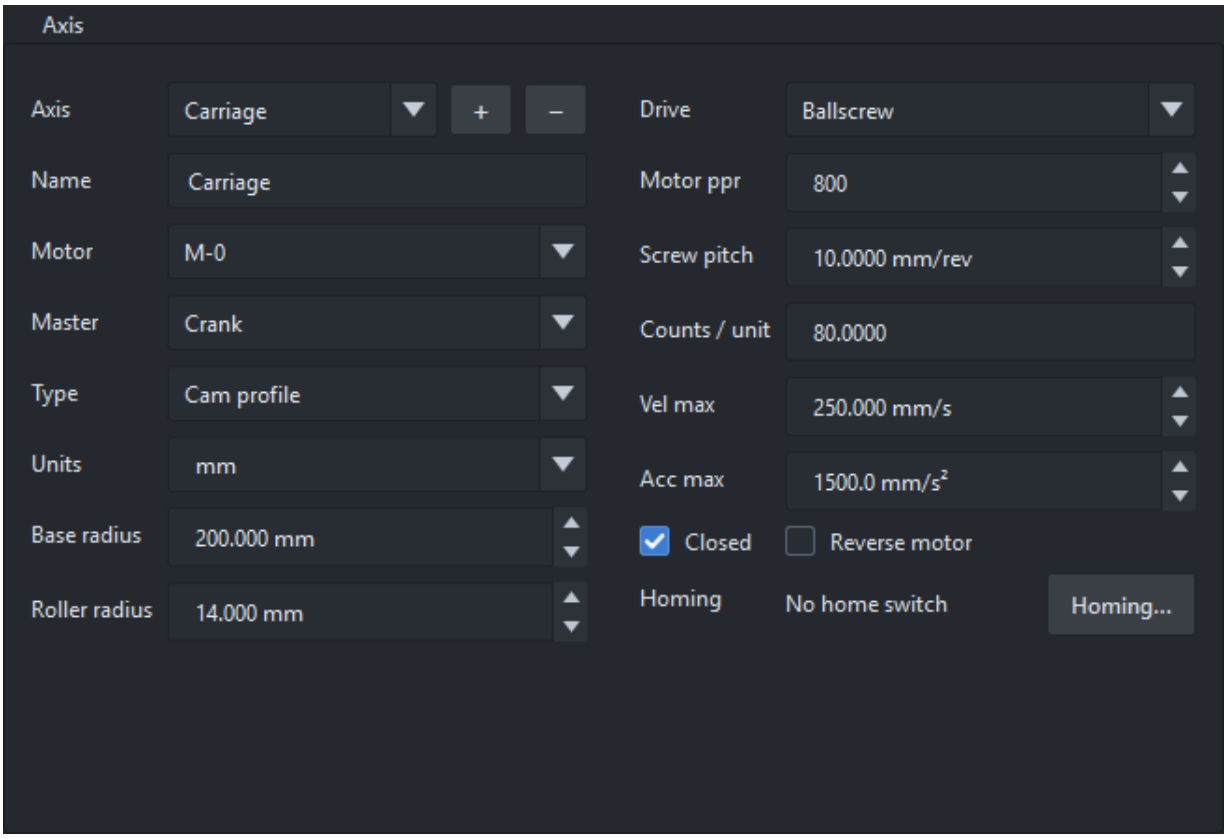


Figure 6.1. The Axis page: name, master, motor connector, mechanics and limits.

6.1 Electronic Gear

Set **Type** to **Electronic gear** for an axis that simply follows the crank at a fixed ratio: a feed roll, a conveyor, a spindle. It has no diagram. **Gear ratio** is the travel, in this axis's units, for a number of crank revolutions:

- **3 : 2** with units *rev*: three motor revolutions for every two of the crank.
- **5 : 1** on a ballscrew in *mm*: 5 mm of travel per crank revolution.
- **-1 : 1**: the same speed as the crank, the other way.

A negative ratio makes the axis travel the other way. If the axis travels the right way on paper but its motor is mounted or set up the other way round, use Reverse motor direction (section 6.6) instead.

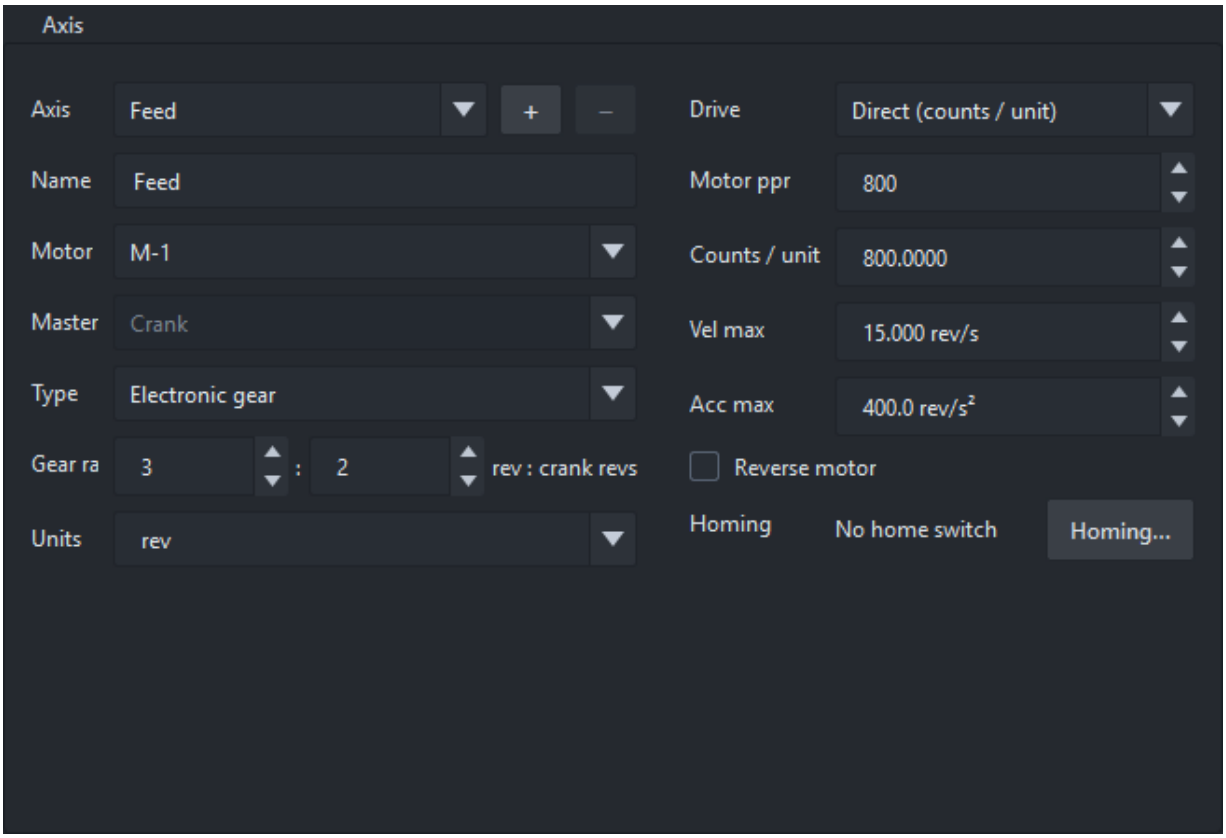


Figure 6.2. An axis set up as an electronic gear: 3 motor revolutions for every 2 crank revolutions.

Both numbers are whole, and the board keeps the ratio exact: after any number of revolutions, in either direction, the axis is right to the count, however awkward the ratio (3 : 7 does not drift). It starts, ramps, stops and jogs with the crank, from an encoder or a driven crank, and the Analysis page shows its speed and limits as for any axis. The diagram shows one revolution of the crank as a straight line, and neither it nor the Segments table can be edited.

A gear always follows the crank, so **Master** is fixed. In cut-length mode (chapter 8) the crank is the line and the ratio is per cycle *as drawn*: changing the cut length leaves the gear's speed against the line as it was, so 100 : 1 on a 100 mm design length is 1 mm per mm of line whatever the cut.

6.2 The Drive

The drive turns positions in your units into motor steps (**counts per unit**):

Drive	Counts per unit	Example
Direct	typed in	a rotary table in degrees: 800 steps ÷ 360° = 2.222
Ballscrew	motor steps per rev ÷ screw lead	800 ÷ 5 mm = 160 counts/mm
Belt	motor steps per rev ÷ (π × pulley pitch diameter)	800 ÷ (π × 25 mm) = 10.19 counts/mm

Units are free text (`mm`, `in`, `deg`, ...); lead and pitch diameter are in the axis's units. Match **motor steps per rev** to the ClearPath's input resolution (MSP).

6.3 Limits

Optional **velocity** and **acceleration** limits (per second). With them set:

- the Crank page gives the highest safe crank speed, and the Timeline page the highest safe override;
- Analysis warns when the profile exceeds them at the set speed (including the extra acceleration while a driven crank ramps).

6.4 Motor Speed

Analysis shows each axis's peak **motor rpm** and **step rate**, and a driven crank's motor rpm. Keep them inside the motor's rating; for a first run on the bench, well below it.

6.5 Cam Drawing

For a crank axis, **base radius** and **roller radius** shape the plate cam in the cam view and the DXF export only. A larger base circle lowers the pressure angle; a roller larger than the cam's smallest curvature radius undercuts (Analysis warns).

6.6 Motor Direction

Tick **Reverse motor** on the Axis page if a motor turns the wrong way, and Upload. For a driven crank the same box is on the Crank page. Set it with the motor free of the machine, or at low

speed, the first time.

Only the motor changes: the diagram, the positions and the Analysis stay as drawn, so positive still means positive travel of the axis. Two motors can also differ because a drive is set up the other way round in its own software; this box covers that too.

6.7 Homing

When the board starts, it doesn't know where an axis is: it takes every axis to be at the profile's start. Homing finds out. The board drives the axis to its **home switch**, catches the exact point where the switch trips, and sets the axis's position there. From then on the axis's positions are real, every time.

It needs a home switch: an input with the **Home** job for the axis, on the Inputs (section 9.9) or CCIO-8 (section 9.10) page. The switch wizard (section 9.12) sets one up and proves it, and can set up homing too. The **Homing** row on the Axis page shows the switch; **Homing...** opens the settings:

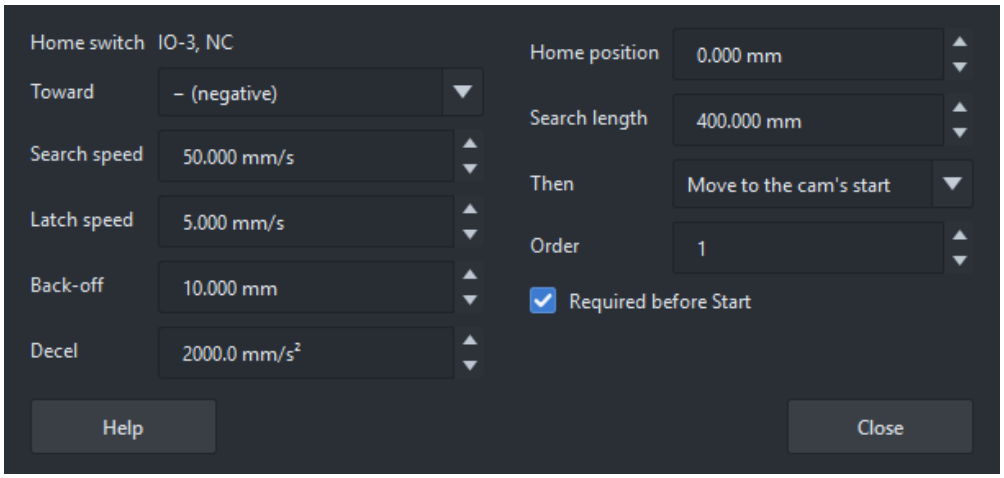


Figure 6.3. The Homing window of an axis: the switch, the speeds, and what to do after homing.

6.7.1 The Sequence

1. **Search.** The axis moves toward the switch at the **Search speed**. (If it starts on the switch, it first moves off it, and the back-off distance more.)
2. **Run-on.** The switch trips and the axis stops, at the **Decel**: it runs a little past the switch while it slows (speed² ÷ 2 × decel).
3. **Back off.** It moves away until the switch releases, then the **Back-off** distance more.
4. **Latch.** It comes back slowly, at the **Latch speed**. The sample the switch trips is the latch: the axis's position there becomes the **Home position**. Coming in slowly, from the same side every time, is what makes the position repeat to a count.
5. **Then** it moves to the cam's start, so Start can go, or stays at the switch.

Limits stay active all the way. Homing fails, with the fault *home failed on X:* ... and the motors still enabled, if a limit trips first, the search runs past its **Search length**, or the switch doesn't release.



Figure 6.4. Homing toward −: the axis's position over time.

6.7.2 The Settings

Toward	Which way the switch is: − or +.
Search speed	The search and the back-off. 10 % of the axis's jog speed to begin with: set it to suit the machine before homing it.
Latch speed	The last, slow move onto the switch: slower repeats better.
Back-off	How far past the switch's release it backs off before the slow move.
Decel	How hard it stops on reaching the switch, after the search and after the latch. The axis runs on past the switch by speed² ÷ (2 × decel): at 100 mm/s and 2000 mm/s², 2.5 mm. *As axis jog* uses the axis's Acc max, or the search speed in a quarter second (100 mm/s: 12.5 mm). Leave that much room between the switch and the hard end.
Home position	The axis's position at the switch's edge, measured like the diagram (the profile's start is 0).
Search length	The longest search: past it, homing fails. Set it a little over the axis's travel. *Until the switch* searches until the switch or a limit; Analysis warns when there is no limit that way.
Then	*Move to the cam's start*, so Start can go, or *Stay at the switch*.
Order	For **Home all**: lower numbers home first, the same number together (a Z axis up before X moves, say).
Required before Start	The board refuses Start until the axis has homed since its motor was enabled.

6.7.3 Running It

Home from Axis jog (section 9.4) (**Home** on an axis's card, **Home all** at its foot), from **Home** in the strip at the foot of the main window (every axis with a switch, in order), from a PLC (the HOME bit), or with `axhome` on the console. **Stop** stops it. An axis stays homed until its motor is disabled, a fault disables the motors, it is zeroed, a new profile is uploaded or the board restarts: a disabled ClearPath can be turned by hand, so the board can't vouch for its position after that. An axis left at the switch (*Stay*) is off its cam position, so Start waits until it is moved there (**To cam position**).

CAUTION

Before homing a real machine, set the speeds, the decel and a search length. A project from before homing existed starts with defaults worked out from the axis's jog limit, which can be far too fast for a machine.

CHAPTER 7

Outputs

The project switches the ClearCore's outputs **IO-0..IO-5** with **pulses**, like the cam switches of a mechanical machine: a saw motor, a clamp, a glue gun, a vacuum valve. Outputs belong to the whole project, and each pulse picks its master:

- **Crank pulse**: on from a to b crank degrees, every revolution; on while the crank is engaged.
- **Timeline pulse**: on from t_1 to t_2 seconds of timeline time; it follows the speed override, repeat and hand jog with the timeline axes.

Up to 32 pulses (at least 0.1° or 1 ms each). **Every output is off** when its master is not running: idle, armed, done, after Stop, Abort, a fault or the estop.

7.1 The Outputs Band

At the bottom of the Displacement Diagram, one lane for each output the project uses (an output with at least one pulse); with none, no band. Each lane shows the pulses on the selected axis's master (crank pulses on a crank axis, timeline pulses on a timeline axis; the other master's pulses are counted, greyed). A new output starts on the Outputs page (section 7.2).

- **Double-click** a lane to add a pulse to that output; **drag** it to move it (also onto another lane); drag an **end** to change on or off; **right-click** to delete. It snaps like the knots (Alt: none).
- The **lamp** by each label shows the output at the cursor; the square, the board's output while it runs this project.

7.2 The Outputs Page

Every pulse as a table (output, master, on, off) to type exact values. **Add** puts a pulse at the cursor on the selected axis's master, on the first output not yet used; the **Output** column moves it to another. Changing a pulse's master converts it at the expected crank speed.

CAUTION

An IO pin is an output or the trigger, not both. The timeline's trigger input (Timeline page) can't be an IO pin the outputs use: the trigger list greys those pins, the Output column greys the trigger's, and a project that has both on one pin gets an error in the review (the board refuses to load it).

CAUTION

Simulate does not switch the pins. The board still runs and reports the outputs (the status line and the lamps), so a simulation never fires a real actuator.



Figure 7.1. A pick-and-place timeline: the X axis's move, with the gripper (IO-0) and vacuum (IO-1) pulses in the band.

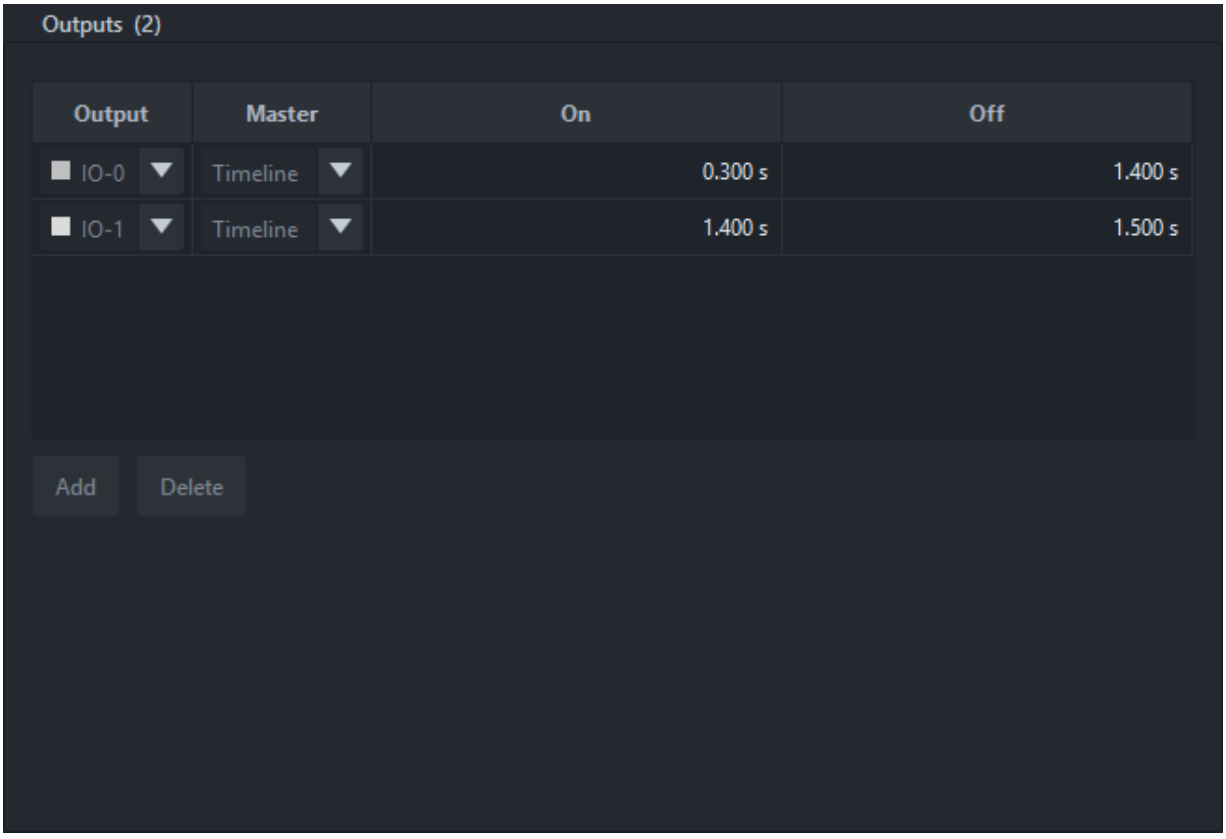


Figure 7.2. The Outputs page: each pulse as a row of output, master, on and off.

CHAPTER 8

Flying Saws: Cut-Length Mode

A flying saw, laser or shear cuts moving material: a carriage catches up with the line, rides it while the cut is made, and returns for the next piece. The piece length usually changes on the fly. Tick **Cut-length mode** in the Crank page. An encoder can measure the line, or a driven crank's motor can feed it.

8.1 Setting It Up

- **Encoder counts / rev** and the measuring **wheel** circumference give the encoder counts per mm of line.
- The profile is drawn over one cut of the **design length**: 360° is the design length, and the diagram's axis reads in mm of line.
- **Line speed** (m/min) replaces the crank speed for the analysis.
- **Kerf**: the material the cut removes (a saw blade; 0 for a shear). A finished piece is the cut-to-cut distance less the kerf.

8.2 On a Driven Crank

With a driven crank there is no encoder: the crank motor feeds the line, and its counts are the line. **Line per motor rev** replaces the wheel: the line one motor revolution feeds, through its gearbox and feed rolls. Line speed, the ramp and the motor max read in m/min, and the motor runs at that speed through every cut, whatever the length. Nothing measures the line, so material slipping in the feed rolls shows up as a wrong piece length. Where that matters, put an encoder on the line. The `rpm` command and setpoint are cuts per minute at the design length.

8.3 What Stretches

Only the **wait** at the start of the cycle (where every crank axis dwells) stretches or shrinks with the length. The rest (accelerate, sync, decelerate, return) keeps its line distance, so speeds, limits and motor speeds are the same at every length: the Crank page gives one **max line speed** for all lengths. Every crank axis must start with a dwell, and crank output pulses come after it.

8.4 The Cut Line

The carriage leaves home while the next cut line is still upstream of the blade, accelerates (a Blend), and reaches line speed exactly on it at the start of the **sync** (the carriage's Linear segment); the blade then rides the cut line through the sync. The **Cut length** row on the Board page shows the live distance from the blade to the cut line, which reads **on the cut line** through the sync.

- **Shortest piece** = design length − wait − kerf (the carriage must be home before the next cut); there is no longest. The Crank page shows it.



Figure 8.1. The carriage: wait (dwell), blend up to line speed, ride the line (linear, shaded green: the sync), blend down, return. The saw plunges inside the sync.

- **On the board**, the Cut length row (or the `length` command) sets the finished length. It takes effect with the **next cut**; below the shortest piece it is refused.

Build one from scratch in Worked examples: a flying saw (section 10.3).

CHAPTER 9

Running on the ClearCore

The ClearCore runs the profile itself from flash (firmware `ECam_Firmware`); the app only talks to it to upload, start, stop and watch. The **board strip** at the bottom of the window runs it, and the **Board** page in the project tree holds the rest.

9.1 Connecting

Open the **Board** page, then:

- **Ethernet**: **Discover** finds the board, or type its IP, then **Connect** in the strip.
- **USB**: pick the ClearCore's COM port, then **Connect**.

9.2 The Board Strip



Figure 9.1. The board strip at the foot of the window.

Always in view, whatever the tree has open. The light is grey when not connected, green when connected, amber when a PLC has control, and red on a fault.

Button	What it does
Connect	Connects to the board chosen on the Board page, or disconnects.
Upload	Sends the project. The board keeps it in flash and loads it at power-up, motors disabled.
Enable / **Disable**	The profile's motors. **Clear fault** appears while one is latched.
Start	Crank axes engage (a driven crank ramps up) and the timeline arms. **Trigger** appears with timeline axes and starts the armed timeline.
Stop / **Abort**	**Stop**: a timeline finishes its cycle and rests at its end, back where the next Start begins; **Stop** again while it finishes ramps it to rest at once. A crank ramps to rest. **Abort** stops at once. After a stop part-way through a timeline's cycle, **Start** waits until its axes are back at their start: **Hand jog** the timeline to 0, or To cam position (section 9.4).

The status beside the light shows the board's state, whether its profile is this project, the outputs that are on, and warns when the profile is not in flash or no estop is set. With a PLC connected over EtherNet/IP it shows *PLC … connected*, or *PLC … in control*: the board then refuses Upload, Enable, Start and the rest, and those buttons go grey; Stop, Abort and Disable still work (see PLC (EtherNet/IP) (chapter 15)).

9.3 The Board Page

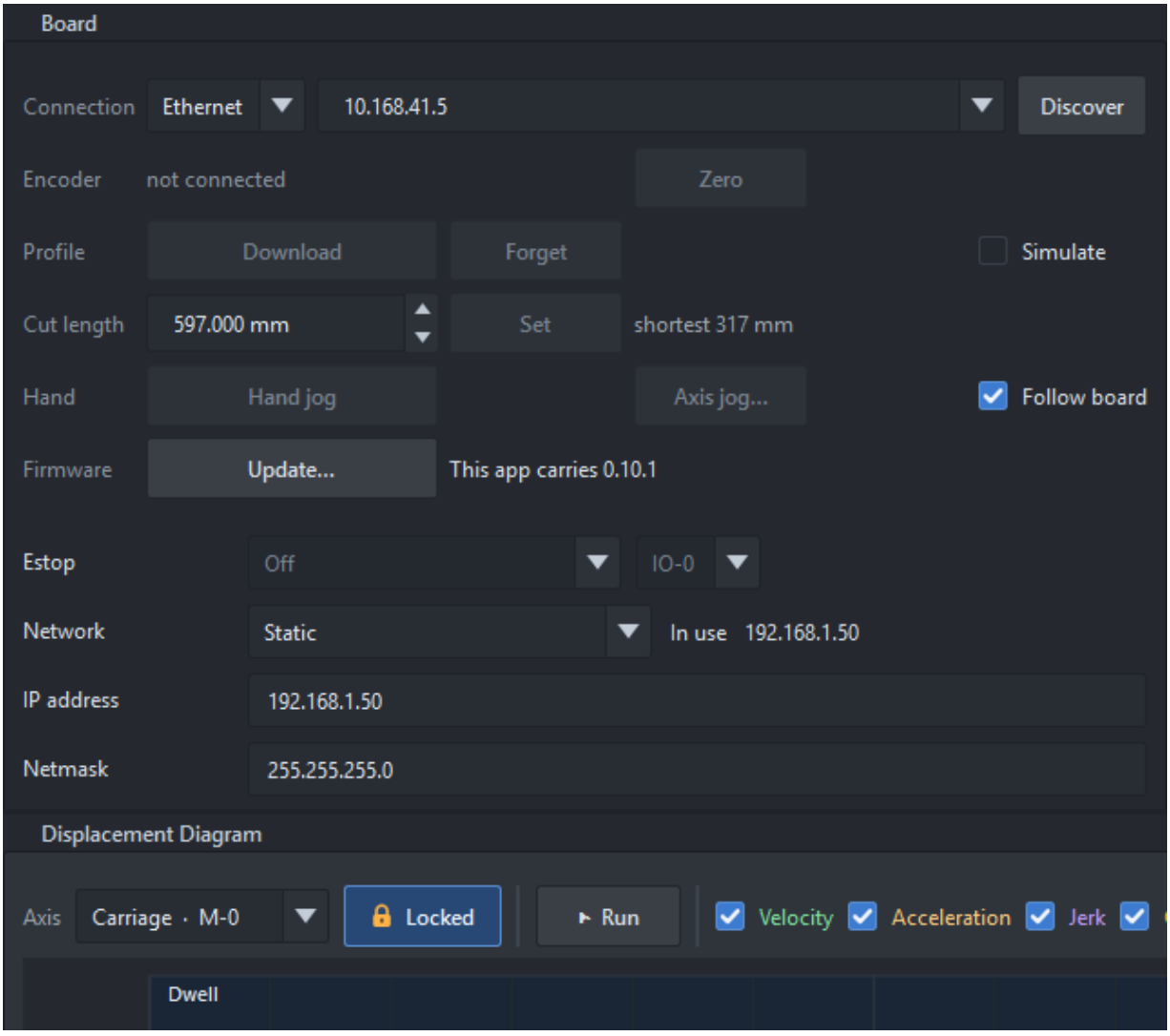


Figure 9.2. The Board page: connection, encoder, profile, hand jog, firmware, estop and network.

Row	What it does
Connection	Ethernet or USB, the address or COM port, and **Discover**.
Encoder	The live count (encoder master), in revolutions and, in cut-length mode, mm of line. **Zero** measures from here (in the app only).
Sim encoder	While simulating with an encoder master: a speed and **Run** / **Stop**; the board follows a simulated count.
Profile	**Download** opens the board's project. **Forget** erases the stored copy. Tick **Simulate** to run without stepping the motors or switching outputs.
Cut length	Cut-length mode: the finished piece length and **Set** (from the next cut); the length being cut, the next and the shortest.
Timeline	**Trigger on Start** makes Start fire the timeline too, and what the timeline is doing.
Hand	**Hand jog**: drag the cam or scrub the diagram and the board's master goes there with every motor following its profile. **Crank → 0°**. **Axis jog...** moves each axis on its own (below (section 9.4)). **Follow board** moves the cursor with the board.
Firmware	The board's firmware against this app's, and **Update...** (see Updating the firmware (section 9.13)).

9.4 Axis Jog

Axis jog... opens a card for each motor connector, M-0 to M-3, with or without a profile on the board. The ClearCore steps the motors, as it does for the cam: the motors' USB isn't used.

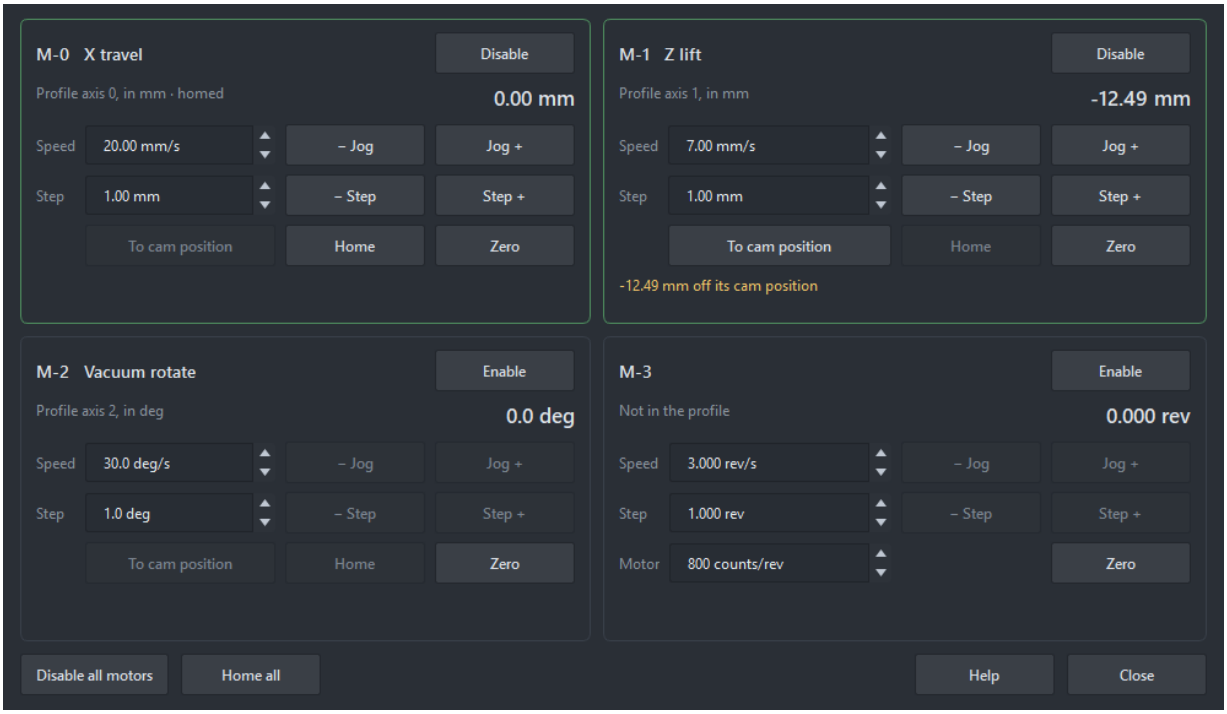


Figure 9.3. X travel enabled and homed; Z lift jogged 12.49 mm off its cam position; Vacuum rotate disabled; M-3 has no axis in the profile.

- A **profile axis** shows its name and moves in its units. A motor **not in the profile** moves in revolutions: set its counts per revolution (as in ClearPath-MSP) on its card. A **driven**

crank's motor only enables: **Hand jog** turns it, with the axes following.

- **Enable** / **Disable** one motor. The others stay as they are.
- Hold **− Jog** or **Jog +** to jog at the **Speed**; let go to stop. If the board stops hearing from the panel (a cable pulled, the app closed) it stops the jog by itself after half a second.
- **− Step** / **Step +** move the **Step** distance, exactly. Steps on several motors run at once.
- **Zero** makes the position here 0 (an axis: the profile's start).
- **To cam position** moves an axis back where the cam has it.
- **Home** homes an axis with a home switch (Homing (section 6.7)); its card says *homed* or *not homed*. **Home all**, at the foot, homes every axis with a switch in their order. While it homes the button stops it.

The cam engages from wherever the motors are. So after an axis has been jogged, **Start** and **Hand jog** wait until it is back on its cam position: its card says how far off it is. Move it back, or zero it where it belongs. If you press Start anyway, the Workbench lists every axis that is off and offers **Move to cam positions**: it moves them back one at a time, M-0 first, each at its Axis jog speed, and leaves Start to you. The motors must be enabled (or the board in simulate). A gear or an open profile has no fixed place and is never held up.

> **NOTE:** Nothing in the panel works while a cam runs, waits on its trigger, or hand jogs, while the PLC has control, or with a fault latched. The panel then shows why, with **Stop** and **Disable all motors**, which always work. The board can't tell where a motor really is: after a power cycle or a new profile it takes every axis to be at the profile's start.

9.5 Estop

The **Estop** row at the top right of the Board page sets the board's estop: **Off**, **Fault when low** (a normally closed estop, or a jumper to 24 V, holds its input high) or **Fault when high**, and its **input**, any of IO-0..A-12 (A-9 unless you pick another). Beside it, what that input reads now, and whether that is OK or open with the setting chosen, so you can check the wiring before turning it on. It saves on the board at once, over USB or Ethernet, with every motor disabled; turning it off asks first. The input list greys the pins this project uses, and the board refuses one its profile uses (an output, the trigger, the Encoder Adapter's DI-6..DI-8 with an encoder master). An open estop disables the motors and latches a fault, and enable is refused until it closes.

9.6 Operator Inputs

A machine run without the Workbench or a PLC needs its own buttons. **Panel › Operator inputs...** on the Board page puts them on the ClearCore's inputs:

Enable	Enables the motors of the job on the board, as Enable.
Disable	Disables them, as Disable. It always works, even while the PLC has control. Wire it normally closed: a broken wire then disables.
Fault reset	Clears a latched fault, as Clear fault.
Start	As Start. With a timeline whose trigger is *Command* it triggers it too: a panel has no Trigger button.

Stop	As Stop: a timeline finishes its cycle; pressed again, it stops at once. It always works. Wire it normally closed.

Each is an input or *Off*, with its contact: **NC** is pressed when the input opens, **NO** when it closes. **Enable** is either *momentary* (an Enable button and a Disable button) or *maintained* (a switch: on enables, off disables). The window shows each input's state, and greys the inputs the job uses, the estop's and the other operator inputs'.

- **Edges, never levels at power-up.** Enable, Fault reset and Start act when the button is pressed, not when it is already held as the board starts: a switch left on doesn't enable a machine by itself; turn it off and on.
- Held, **Disable** keeps Enable from acting and **Stop** keeps Start from acting.
- A press must last 10 ms: contact bounce doesn't count.
- While the **PLC has control**, Enable, Fault reset and Start are the PLC's: the panel's are ignored. Otherwise the panel and the Workbench both work.

These are the board's settings, like the estop: they stay with the board whatever job it runs, and change only with the motors disabled and nothing moving. The board refuses a job that uses an operator input's pin, and the Workbench keeps those pins off the Inputs page (*Operator: Enable*), the trigger list, the Outputs and Waits pages and the switch wizard, remembering them from the last board it saw.

9.7 Network Settings

The right of the Board page shows the board's network settings once connected, and the address it is using now. To change them, connect over **USB**:

1. Choose **DHCP** or **Static**, and set the IP address, netmask and gateway (0.0.0.0 for none). On DHCP the address is the **fallback**: the board uses it if no server answers, which takes about 13 seconds.
2. Leave **Host name** empty for the default, `ecam-` and the last 3 bytes of the MAC.
3. **Apply and restart**. Network settings only take effect when the board restarts, so the app restarts it, reconnects and reads them back. *Applied.* means the board kept them.

They can be changed with the motors disabled, nothing moving and no PLC in control. The app checks them first: a bad address, a mask with holes or a gateway on another subnet can't be applied, and the reason shows beside the button.

> **NOTE: Why USB only:** a new address would take away the Ethernet link it was set on. USB reaches a board whatever its address, so it is also the way back to a board on the wrong network. A PLC or EtherNet/IP tool can change them too (see PLC (EtherNet/IP) (section 15.7)).

CAUTION

The restart drops any PLC connection. It comes back by itself if the board keeps the address the PLC is set to; otherwise change the PLC's setting to match.

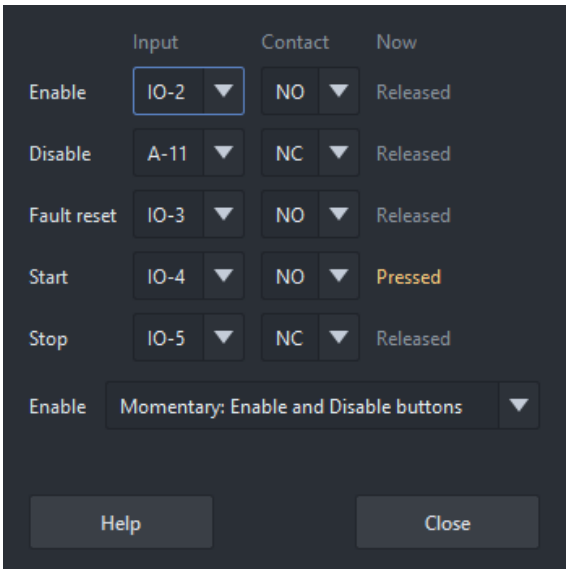


Figure 9.4. A panel with all five; Start is being pressed. Disable and Stop are wired normally closed.

9.8 A Safe First Run

1. **Upload**; the status says *profile … = this project*.
2. Tick **Simulate**, **Start**: watch the cam and the diagram follow (nothing moves, no outputs switch).
3. **Stop**. Untick Simulate, **Enable**; check the motors hold.
4. Use **Hand jog** to move slowly through the cycle and watch the machine.
5. Then **Start** at a low speed, and raise it towards the speed Analysis allows.

CAUTION

The estop. Until one is wired and set, the status line says *no estop*. Set it with Estop (section 9.5) on the Board page.

9.9 Inputs

Inputs in the project tree sets up the ClearCore's own inputs for homes and limits, the same way as the CCIO-8 inputs (section 9.10): **IO-0** to **IO-5**, **DI-6** to **DI-8** and **A-9** to **A-12**. Give an input a **Name**, a **Function** (*Home*, *Limit +* or *Limit −*), its **Axis** and its **Contact** (*NC*, normally closed, or *NO*, normally open). Connected to the board, **State** shows each *On* or *Off*. The limits act on the board: see Limits (section 9.11); a home switch homes its axis: see Homing (section 6.7).

Set up switches... at the top opens the switch wizard (section 9.12).

A pin can do one job, so some can't be inputs:

- an IO pin with an output pulse (*Output*), and an input's pin is not offered on the Outputs page (section 7.2);
- the timeline's trigger input (*Timeline trigger*), and an input's pin is greyed in the trigger list;

- DI-6 to DI-8 while the crank follows an encoder (*Encoder Adapter*). The other way round, with any of them set up as an input, **External encoder** can't be picked on the Crank page.

The board's estop input can't be anything else: the Inputs page shows it as *Estop*, and the trigger list and the Outputs page don't offer it. The board says which it is when connected, and the app remembers it, so the checks hold offline too (*Estop* (*last seen*)). The estop can't move onto an input the board's profile uses. An axis can have one home and one limit each way, counting the CCIO-8's and these together: the same job on two inputs is marked in the tree.

9.10 CCIO-8 Inputs

CCIO-8 expansion boards chain from **COM-1**, up to 8, for the Home and Limit switches. Their limits act as the board's own do (Limits (section 9.11)).

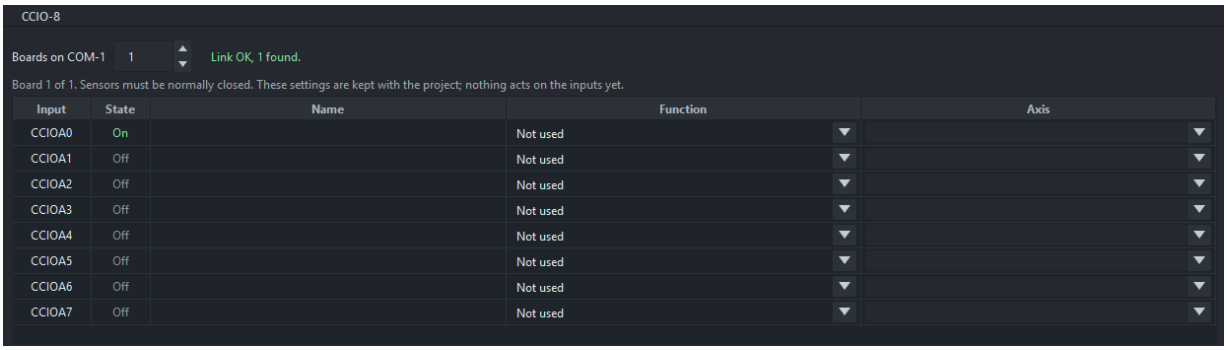


Figure 9.5. The CCIO-8 page: the boards on COM-1 and their inputs.

1. Open **CCIO-8** in the project tree and set **Boards on COM-1**. Each board gets a node under it.
2. For each input in use, give it a **Name** and a **Function** (*Home*, *Limit* + or *Limit* −), pick its **Axis** and its **Contact**. These are kept with the project. An axis can have one of each: the same job on two inputs is marked in the tree.
3. Connected to the board, **State** shows each input *On* or *Off*, about twice a second, and the line at the top says whether the link is good and the board found the number of boards the project has.

> **NOTE: Use normally closed limits.** A normally closed switch reads *On*, and a tripped one or a cut wire reads *Off*, so a broken wire stops the axis. A normally open limit is allowed, but a cut wire reads as not tripped and the limit is lost without a sign: Analysis warns about each one. A broken CCIO-8 link trips every limit on it.

The board looks for CCIO-8s when it starts, so restart it after adding one. If the link breaks while it runs, the page says so in red and every input reads *Off*, as a tripped sensor would, until it comes back. The ClearCore's lights flash the CCIO error meanwhile. It comes back by itself when the cable is plugged in, and the lights stop.

The profile on the board decides whether the link is used. With no profile the link stays up, so you can see the inputs before making a project. A profile uploaded with 0 boards turns the link off: the page still shows the boards found, the inputs show a dash, and a board that drops out raises no error. Upload a project with boards to turn it on.

9.11 Limits

A limit stops the machine under control: quickly, but on a ramp, so the axes stay in step and nothing is thrown. It acts only when its axis moves toward it, so the axis can always be driven off it.

- **The cam running.** The crank or the timeline ramps to rest over the **Limit stop** time (Inputs page, 0.1 s by default), and every axis follows it. With an external encoder as the master, which can't be slowed, each axis lets go and ramps to rest by itself over the same time. The status line shows *limit + on Slide* (*IO-3*), and the motors stay enabled.
- **Clear** it, then use Axis jog (section 9.4) to move the axis off the switch: the jog button toward the limit is greyed while it is tripped. Start is refused while an axis is on a limit.
- **Axis jog** toward a limit that trips just stops that axis; nothing faults.
- **Simulate** only shows a limit tripping; nothing stops.

The axis travels on past the switch while it ramps down. **Overtravel at peak speed**, beside Limit stop, is how far that is for the fastest axis at the profile's top speed: leave at least that much room between each limit switch and the hard end. A shorter stop time travels less, but stops harder. The Status panel under the project tree shows *Limit +* or *Limit −* for an axis on one.

9.12 The Switch Wizard

Tools › Set up limits and homing... (or **Set up switches...** on the Inputs page) sets up one axis's switches from the wiring, and proves each one by moving the axis to it. Connect to the board and upload the project first.

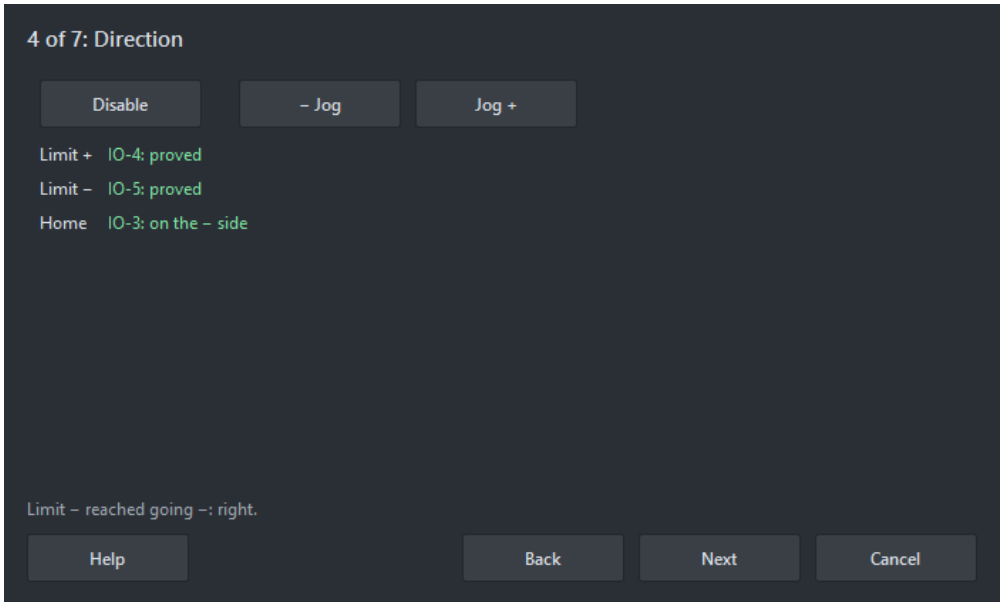


Figure 9.6. The Direction step: both limits reached the way their jobs say, the home switch seen on the − side.

1. **Axis.** Pick the axis, tick the switches it has and set a slow **Setup speed**.

2. **Find each switch.** Trip each switch by hand when asked, and let go: the input that changed is that switch's. An input that is already an output, the trigger, the estop or another axis's switch is turned down.
3. **Contact.** Each switch's level at rest gives its contact (*On* at rest: normally closed). Check it.
4. **Direction. Enable** the motor and hold **Jog +** or **Jog −**. The board stops the axis at the first switch it reaches, and that proves the switch: the + limit must be reached going +. If the axis reaches the other limit, the wizard asks whether it moved toward the end where you tripped that limit by hand. *Yes*: the switches were the other way round, and it swaps them. *No*: the motor runs the other way, and it reverses the axis's motor and uploads. A home switch on the way is noted, with its side.
5. **Limits** (an axis with limits). **Next** from Direction uploads the switches, after asking, and the board's own limits take over. **Enable** (the upload turned the motor off), then hold **Jog +** until the + limit stops the axis, and jog back off it; the same with **− Jog**. The wizard shows how far past each switch the axis ran at the setup speed, the travel between the switches, and how far a limit stop runs at the profile's top speed.
6. **Home** (an axis with a home switch). The direction is the side the switch was seen on; the speeds start from the setup speed, or the axis's own if it had homing set up; the search length is a little over the travel the Limits step measured. Set the **Home position** and the rest (as in Homing (section 6.7)), then **Home 5 times**: it uploads, homes the axis that many times, and shows the spread of the positions the switch latched at. A failure stops it and says why.
7. **Review. Apply** writes the switches and the homing settings into the project (one undo step) and uploads it.

9.13 Updating the Firmware

The app and the board's firmware go together, and the app carries the firmware it needs. When they differ, the Firmware row and the status line say so in amber. **Update...** on the Board page, or **Tools ▶ Update board firmware**, puts it on over USB. It also installs a new ClearCore.

1. Connect the ClearCore's USB cable: the firmware only goes on over USB, even when the app is connected to the board over Ethernet. If the app is connected over USB, it lets go of the port when this window opens, so the update can use it. If it is connected over Ethernet, it stays connected until you press Update. Either way it reconnects by itself when the window closes.
2. Pick the port. Only ClearCores are listed, not ClearPath motors on USB. The window says what is on the board and what will happen to its profile.
3. **Update**. It takes about 10 seconds; leave the cable in and the board powered. The window stays open until it has finished.

The board has	What happens
ECam firmware and a profile	The profile is saved to this PC, checked against the board's hash, and put back afterwards. The motors stay disabled.
ECam firmware, no profile	The firmware goes on; there's nothing to save.

The board has	What happens
Other firmware, or a new board	The firmware goes on and replaces what is there. Upload a project afterwards.
Only its bootloader	The same. The IO-0 to IO-5 lights walk while it's in the bootloader. A board that won't show up: double-tap its reset button to start it, then **Refresh**.

It can't start with the motors enabled, the board running, or a PLC in control. A connected PLC loses its connection while the board restarts and comes back by itself. Network settings and the other board settings stay as they are.

> **NOTE: If an update stops part way**, the board may have no working firmware and vanish from USB, but its bootloader isn't touched and the profile stays in a file on this PC until the board has it back. Double-tap the reset button so the IO-0 to IO-5 lights walk, press **Refresh** and update again: the window offers *Put back profile* …, ticked only if the file is surely this board's.

Without the app: `ECam_Firmware.uf2` is in the `firmware` folder of the install. With the board in its bootloader it shows as a USB drive called `CLEAR_BOOT`; copy the file onto it and the board restarts with the new firmware.

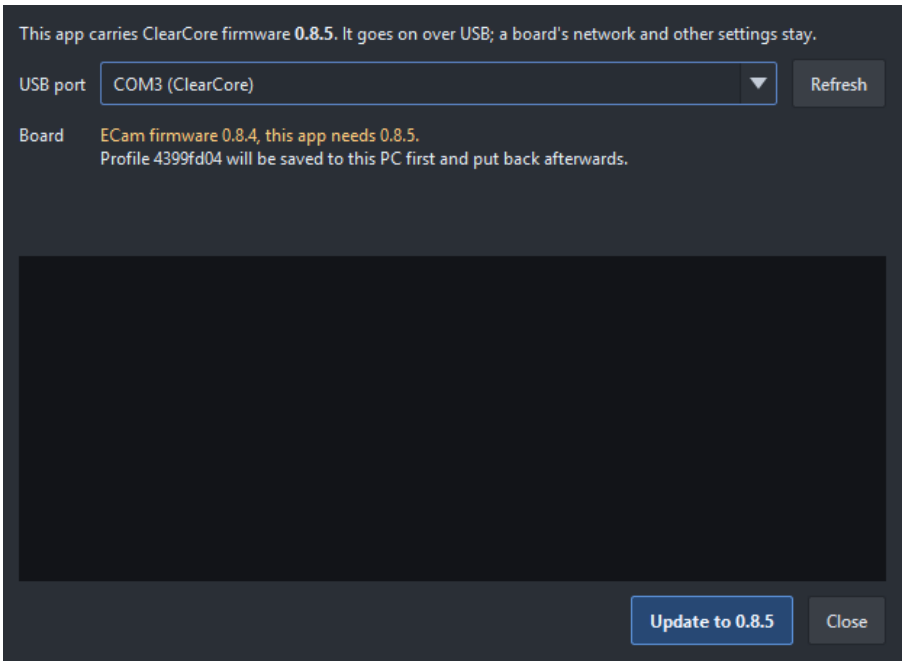


Figure 9.7. Updating the board's firmware over USB.

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Chapter 10

Worked Examples

Four real machines, each built from a blank start. Numbers are a sensible start for a bench test; change them for your machine.

10.1 A Carton Pusher (Rise, Dwell, Return)

The machine: a pusher on a 10 mm ballscrew pushes a carton 50 mm off a conveyor once per machine cycle, holds it while the next one arrives, and returns. An encoder (1000 lines) on the machine's main shaft gives the cycle.

1. Open **File › New from guided setup...** and pick **Rise, dwell, return**.
2. **Machine cycle**: *an encoder on the machine's shaft*; counts per rev **4000** (4 × 1000 lines); expected speed **30 rpm**.
3. **Motor**: name *Pusher*, connector **M-0**, units **mm**, drive **Ballscrew**, 800 steps per rev, lead **10 mm** (80 counts/mm).
4. **Motion**: law **Cycloidal**, stroke **50 mm**; rise 0–120°, return 180–300°.
5. **Create project**, then **File › Save**.



Figure 10.1. The pusher: cycloidal rise and return, with dwells. Velocity and acceleration start and end at zero: no knocks.

Check: Analysis shows 150 mm/s peak at 30 rpm and the motor at 900 rpm. To run faster, set the axis's velocity and acceleration limits: the Crank page then shows the highest safe speed.



Figure 10.2. The same motion as the plate cam it replaces.

10.2 An Index Conveyor

The machine: a belt conveyor (50 mm pulley) advances 150 mm each cycle, then stops while a station works.

1. Guided setup, **Index feed**; encoder or driven crank as for the pusher.
2. **Motor**: drive **Belt and pulley**, pitch diameter **50 mm**.
3. **Motion**: law **Modified sine** (lower peak speed for the belt's load), feed **150 mm** from 0° to 180°.

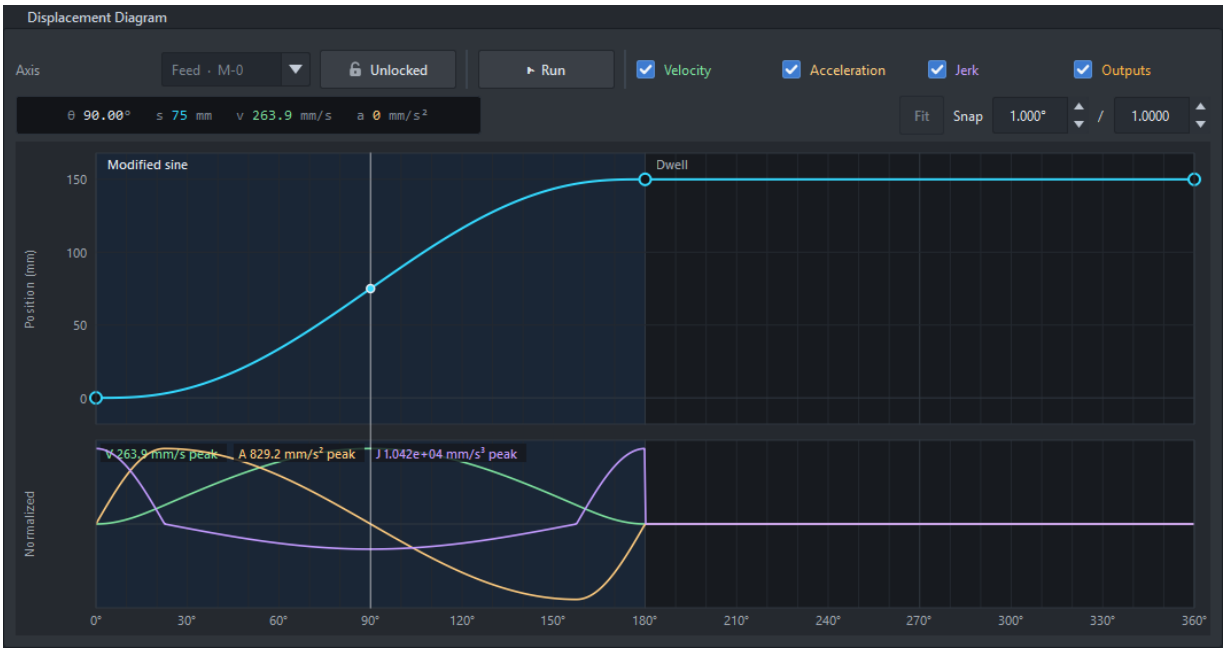


Figure 10.3. An open profile: it ends 150 mm on from where it started, and the next cycle carries on from there.

Check: the profile is *open* (Axis page), so the position keeps adding up cycle after cycle. Analysis shows the advance per revolution.

10.3 A Flying Saw

The machine: extruded profile runs at 5 m/min under an encoder wheel (200 mm circumference, 4000 counts). A carriage on a 10 mm ballscrew rides the line while a saw (second motor, 5 mm screw) plunges 6 mm. Pieces of 600 mm, 3 mm blade. The saw motor (IO-0) runs during the cut.

1. Guided setup, **Flying cut-off**.
2. **Line and cut**: counts per rev **4000**, wheel **200 mm**, cut length **600 mm**, kerf **3 mm**, line speed **5 m/min**.
3. **Carriage motor**: **M-0**, ballscrew, lead **10 mm**.
4. **Saw motor**: **M-1**, ballscrew, lead **5 mm**; keep *Switch output IO-0 on while riding the line*.
5. **Motion**: catch up in **36°**, ride for **36°**, return in **78°**, plunge **6 mm**.

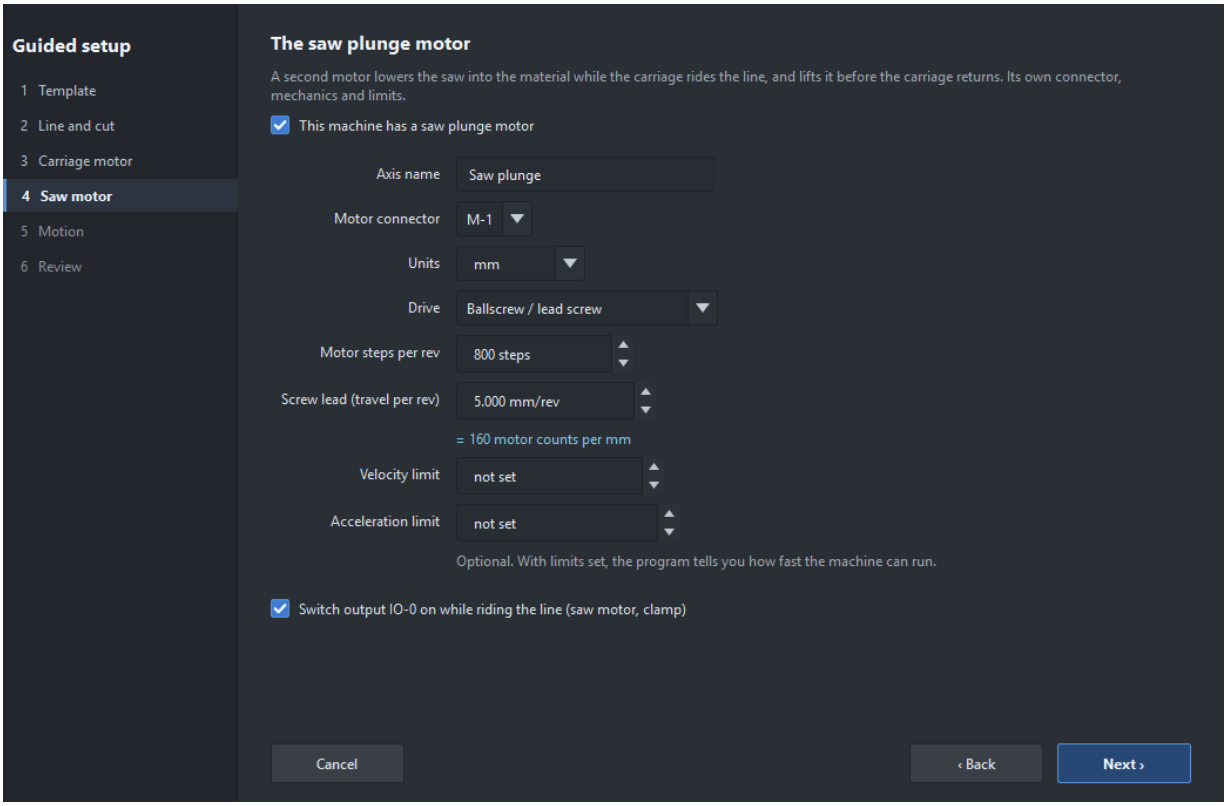


Figure 10.4. The saw motor has its own step: its connector, mechanics and limits.

Run it: upload, tick **Simulate**, **Start**, then **Run** the **Sim encoder** at 5 m/min. Change the **Cut length** on the Board page while it runs: the next piece takes the new length. See cut-length mode (chapter 8).

10.4 A Triggered Transfer (Timeline)

The machine: a belt-driven transfer moves a part 200 mm when a sensor on **IO-0** sees it, holds it 0.4 s, and comes back. No machine shaft: the timeline times it.



Figure 10.5. The flying saw's carriage: wait, blend up to line speed, ride the line through the sync while the saw (IO-0) is on, blend down and return.

1. Guided setup, **Triggered move**.
2. **Trigger**: a digital input, **IO-0**, rising edge; one run per trigger.
3. **Motor**: belt, pitch diameter **25 mm**.
4. **Motion**: law **Modified trapezoid**, distance **200 mm**; out in **0.6 s**, hold **0.4 s**, back in **0.6 s**.

Add a gripper: double-click lane IO-1 in the Outputs band at 0.6 s and drag its end to 1.0 s: the gripper closes for the hold. The **Pick and place** example (**File › Open example**) has three axes and two outputs done this way.

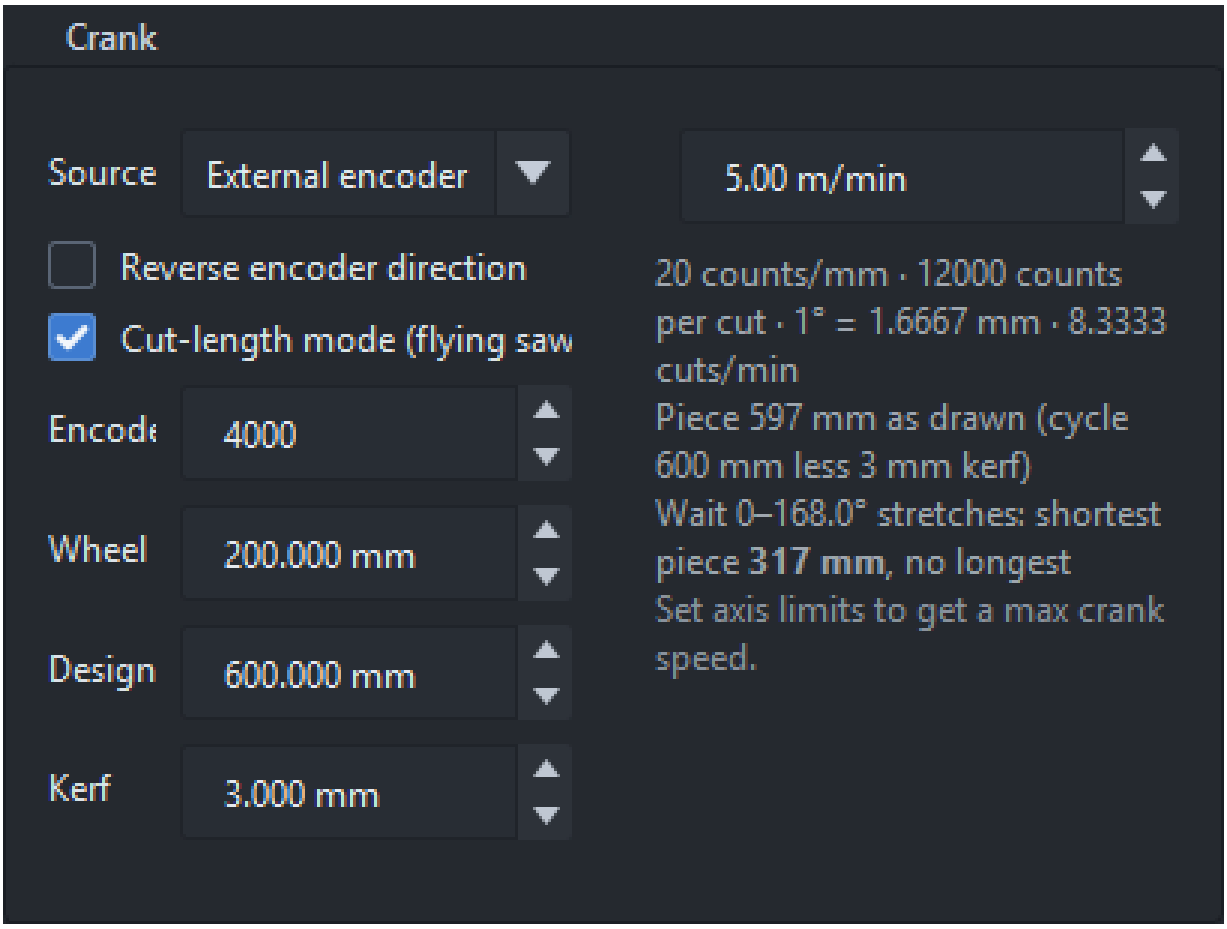


Figure 10.6. The Crank page in cut-length mode: counts per mm, the wait that stretches, and the shortest piece.

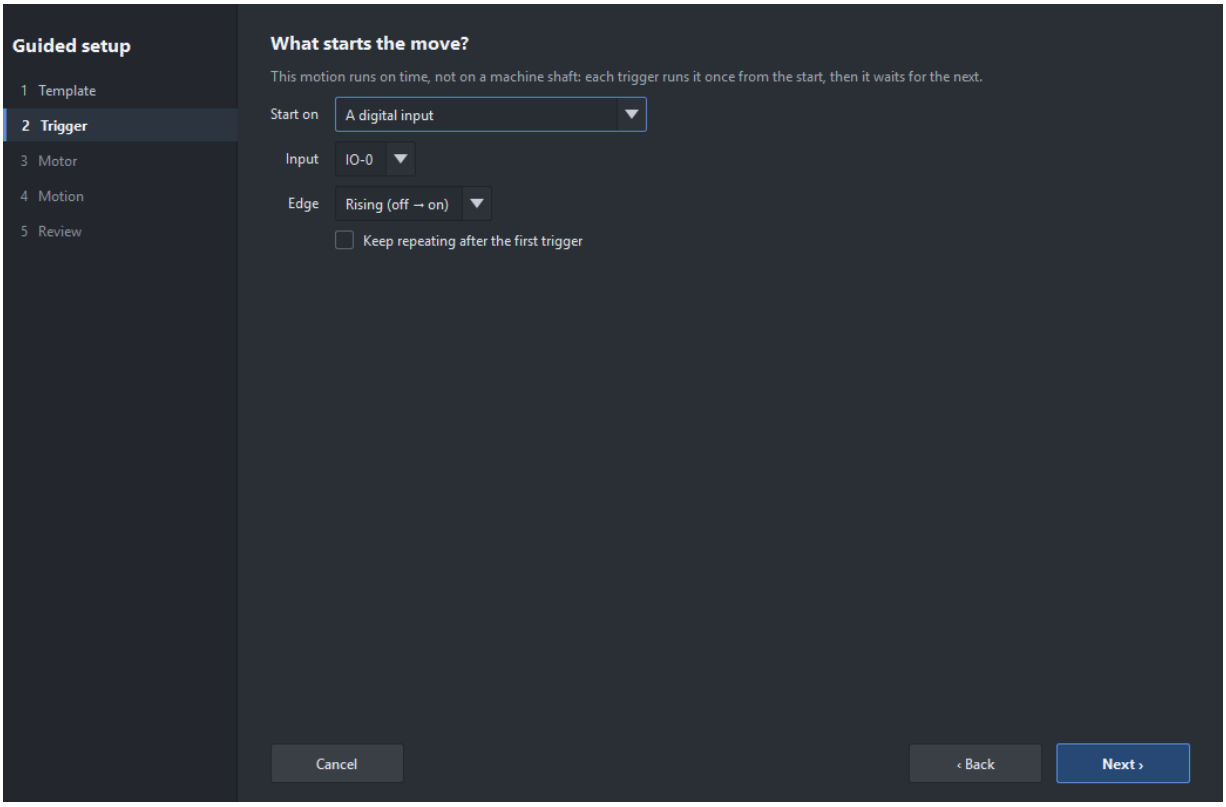


Figure 10.7. Guided setup for a triggered move: the trigger step, here a rising edge on IO-0.



Figure 10.8. The timeline move: out, hold, back; it rests at the end until the next trigger.

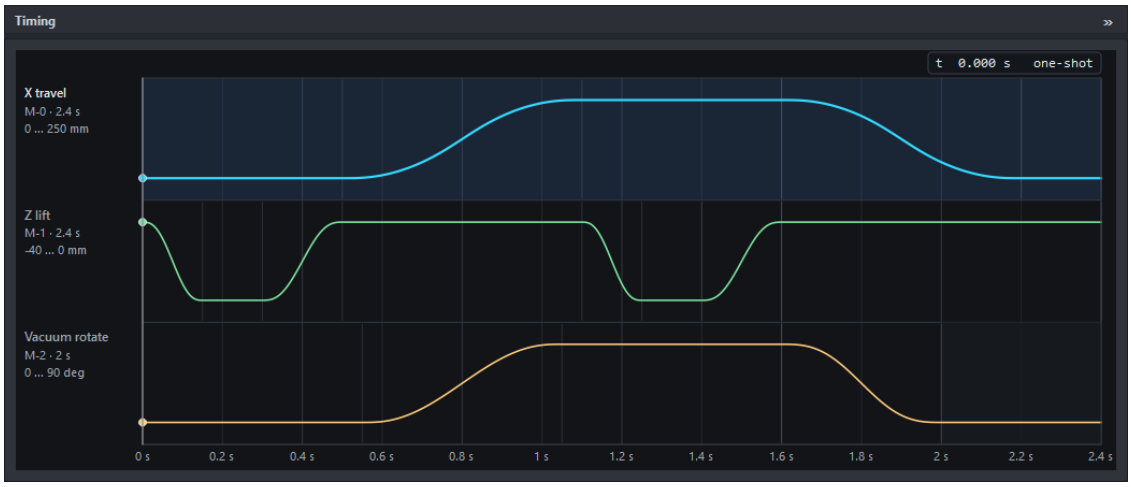


Figure 10.9. The Pick and place example in the Timing view: X travel, Z lift and a vacuum rotate on one timeline.

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CHAPTER 11

Files and Exports

11.1 The Project

A project is one `.ecam` file (JSON): the master, every axis, the outputs, the mechanics and the drawing. **File** › **Save** / **Save as**; **Open recent** lists the last ten.

> **NOTE: The project travels with the upload.** Uploads carry a compressed copy of the whole project, so **Download** gives back exactly what was uploaded, mechanics and limits included.

11.2 Exports

File menu	Writes
Export table (CSV)...	The selected axis sampled over one cycle: master position, master counts, position, motor counts, velocity, acceleration.
Export table (C header)...	The same as an `int32_t` table of motor counts, for a first firmware experiment.
Export cam DXF...	A crank axis's plate cam (DXF R2000, in mm or inches): outline, pitch curve, base circle, optional bore, marks and notes, each on its own layer.
Export path DXF...	The current axis's path (DXF R2000, in mm or inches): its position over one cycle as a polyline, with axes, ticks and notes. Across: on a timeline axis, so much per second (40 mm/s, or 10 in/s, to begin with); on a crank axis, a scale picked so the drawing is about twice as wide as it is tall, at an easy value such as 1 deg = 0.1 mm. Either is written in the notes. Also in the Axis menu and an axis's right-click menu in the project tree.
Export workbook (Excel)...	The whole project as one `.xlsx` with tables and charts (below).
Generate report... (Ctrl + R)	A PDF for reviewing the project (below); it opens when written.

> **NOTE: DXF units:** both DXF exports draw lengths in mm or inches, converted from the axis's units, and set the drawing's units to match. An axis in degrees, revolutions or counts is drawn as it is, unitless. A path DXF's notes also carry a line beginning `ECW-PATH` with its scales, for reading the path back.

11.3 The Excel Workbook

One workbook, ready to share or archive:

- **Summary**: every check, the axes at a glance (stroke, peaks, motor rpm, step rate), the master, links to each sheet.
- **Master**: the master settings, the speed the project allows and why, the output pulses.

- **Timing**: crank and timeline axes overlaid, and the output lanes, as charts.
- Per axis: **Profile** (settings, segments with their peaks, analysis), **Motion** (sampled position, velocity, acceleration, jerk, with a chart of each) and, for crank axes, **Cam** (outline to scale and pressure angle).

11.4 The Project Report

A PDF for the engineer who reviews the project before it runs, laid out for print (Letter or A4 by your region) and drawn as vector graphics, so it stays sharp at any zoom:

- **Cover**: the review status (errors and warnings), key figures (crank speed and headroom, cycle time, axes, outputs, peak motor speed) and the contents.
- **System overview**: a block diagram of the master, the ClearCore, each axis and the outputs, and which motor connector and pin does what.
- **Timing diagram**: every axis on its master, the output pulses below.
- **Machine cycle**: a flow chart of the cycle, one step for each stretch where nothing changes: which axes move (from, to, law), which hold, which outputs are on, and how long it takes.
- **Master**, then **per axis**: settings beside results, its checks, the segment table, the displacement diagram with velocity, acceleration and jerk (limits and peaks marked), and for a crank axis the plate cam and its pressure angle.
- **Outputs**, every **check**, and a **review and sign-off** page with a checklist, notes and signature lines.

The figures and checks are the same as in the app and the workbook.

11.5 Command Line

`ECamWorkbench project.ecam`	open a project (skips the start page)
`--profile-lines project.ecam out.txt`	the board upload and its hash
`--cam-dxf project.ecam <axis> out.dxf`	the plate cam of a crank axis
`--workbook project.ecam out.xlsx [samples]`	the Excel workbook
`--report project.ecam out.pdf`	the project report
`--no-start`	skip the start page

Chapter 12

ClearPath Scope

Tools › ClearPath scope... (Ctrl + Shift + O) records a ClearPath MC/SD through the motor's own USB port (the port ClearPath-MSP uses) and shows its signals stacked on one time axis: what the motor actually did, next to what the cam asked for.



Figure 12.1. The default lanes, the critical concerns: tracking error, velocity, bus voltage, drive temperature, HLFB, enable and alerts.

12.1 Getting Connected

1. Plug the motor's USB into the PC. It shows in **Motor** as *ClearPath on COMn*.
2. **Close ClearPath-MSP** first: only one program can own the port.
3. **Connect**, then **Start**. **Stop** ends the recording; **Disconnect** releases the port.

With more than one motor on USB, the list shows each by its COM port. Connecting shows the drive's serial number, so you can tell which motor it is.

12.2 Several Motors at Once

Up to four motors record together on one time axis, each through its own USB port. **Connect all** connects every ClearPath in the list, or pick one and **Connect**, then pick another and **Add**. **Start**

records them all; **Disconnect** releases them all.

- A **Motors** row appears with a box for each. Type a name, such as *M-0 Carriage*: it shows on that motor's lanes, and stays with the drive by its serial number, so it survives a change of COM port. The app can't tell which connector a drive is on, so the name is how you know.
- Each motor gets its own lanes in **Lanes**, beside the same signal on the other motors. Only its fast signal starts on. Each motor streams one fast signal; a second one is read in turn, more slowly.
- **Overlay** draws two motors in one lane, and the trigger can watch any motor's lane and capture them all.
- **Save CSV** writes every motor into one file, with its name on each signal.

> **NOTE:** The drives share no clock, so samples are lined up by when they reach the PC. On the bench that held to about 1 ms between motors, with no drift over two minutes. That shows how motors move together; it is not for measuring a delay of a millisecond or two between them.

> **NOTE:** No motor? Pick **Simulated motor**: it speaks the same protocol and runs a repeating motion.

12.3 What the Motor Can Give

Kind	Rate	Signals
Streamed	about 845 / s	one monitor variable at a time: position error (default), velocity, torque, torque utilization, bus voltage
Registers	about 10 / s each	velocity, position error, measured position, bus voltage, drive temperature
Digital	about 845 / s	HLFB and Enable, carried in every stream sample
Alert	about 2 / s	whether the alert register is non-zero; the names show in the status row

The motor streams **one** fast signal, so only one "stream, fast" lane can be on. Position error is in encoder counts and velocity in RPM; other streamed signals are in raw monitor units. The bus voltage scale, HLFB and Enable are provisional (from our USB captures).

12.4 Lanes

Lanes opens the lane settings: tick a signal to show it, **Up** / **Down** for the order, click the swatch for its colour, and **Auto** or **Fixed** scale with min and max. Everything is saved.

12.4.1 Overlay

Overlay draws a signal in the lane above it. Signals in one unit share a scale; otherwise each keeps its own, with its scale labels in its colour.

12.5 Reading the Traces

- **Hover**: every lane shows its value at that moment, and the time axis the time.

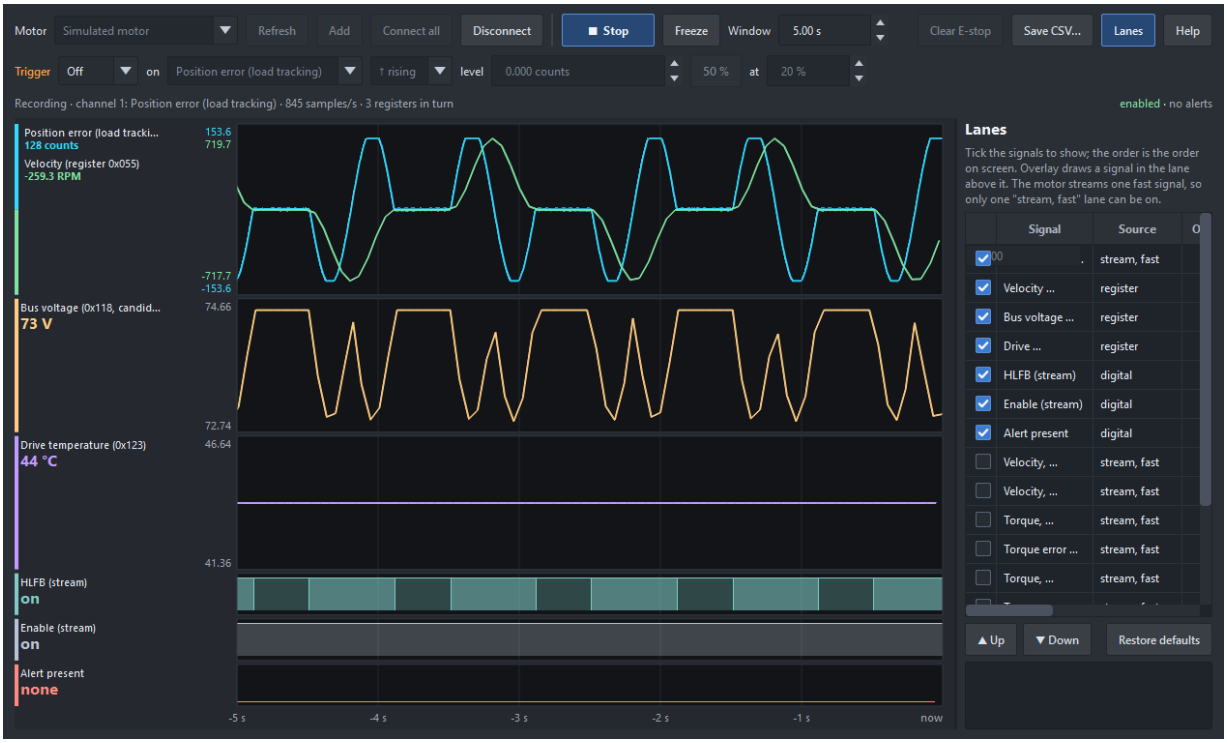


Figure 12.2. The Lanes settings: which signals show, their order, colour and scale.

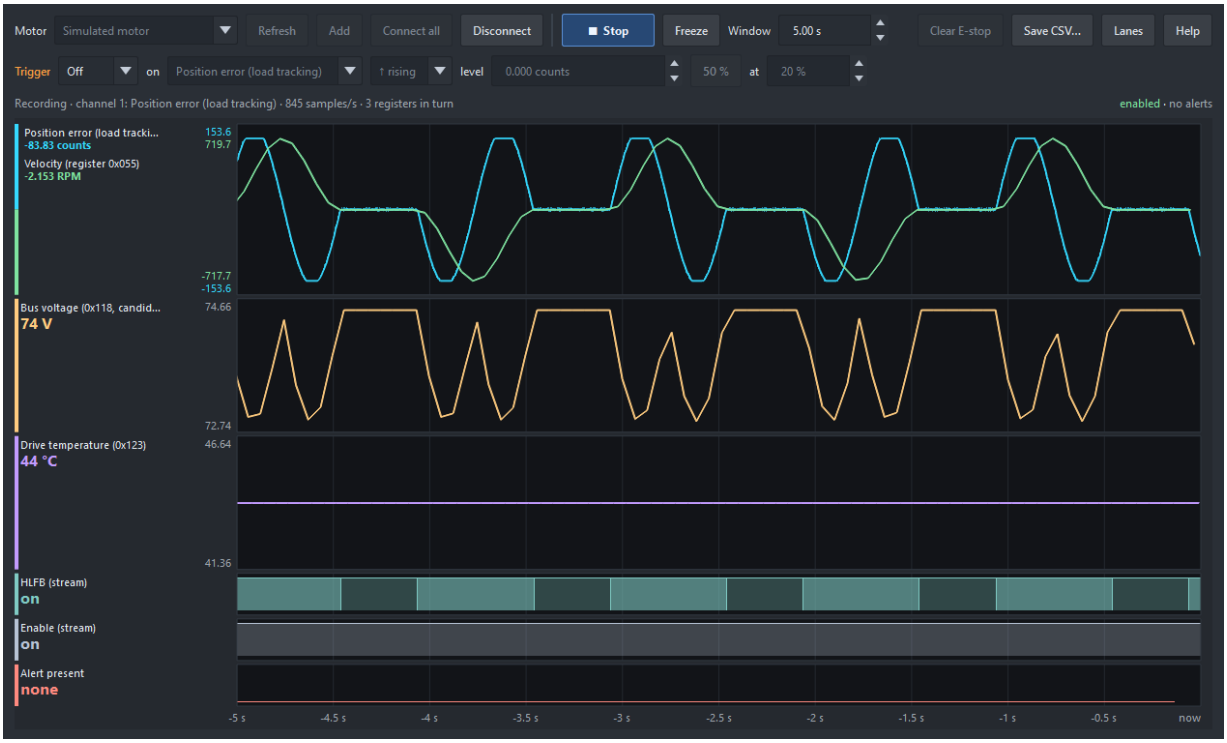


Figure 12.3. Velocity overlaid on the tracking error: the error rises with each acceleration.

- **Freeze** holds the view while recording carries on (a badge shows in the status row).
- **Save CSV...** writes everything recorded (the last two minutes): time, signal, value, unit.

12.6 Zoom and Pan

Zoom changes the **timebase** (the **Window** box: seconds across, 10 ms to 120 s). Stopped or frozen it zooms about the mouse; live, the right edge stays on the newest data.

Ctrl +wheel, Ctrl + / −	zoom
middle-drag, Shift +wheel	pan through time (a live view freezes)
Ctrl + 0	fit everything recorded

12.7 Trigger

A trigger holds the screen on a repeating moment, so each machine cycle stands still for comparing.



Figure 12.4. Normal trigger on the tracking error rising through 0, at 20 % of the screen: the dashed orange line is the moment, the level is marked in its lane, and the time axis reads from the trigger (T).

Off	roll, as without a trigger
Auto	hold each capture; roll live when nothing triggers for a sweep
Normal	hold the latest capture and re-arm after each
Single	one capture, then wait for **Arm**

Choose the **source** (any lane that is on, HLFB and Enable too), the **edge** and the **level** (**50**

% puts it in the middle of the source's range on screen). **at** is where the trigger sits across the screen. Noise at the level cannot re-trigger (2 % hysteresis). Change a setting while stopped and the whole recording is searched again.

Example: to see the tracking error of every move of a pusher, trigger **Normal** on **HLFB**, **↓ falling** (HLFB drops when a move starts), at **10 %**, with a timebase a little longer than the move.

12.8 What the Scope Does to the Motor

Start sets the motor's **working** monitor channel 1 to the streamed signal and speeds the link to 230400 baud; Stop puts channel 1 back exactly as it was and returns the link to 9600 baud. Nothing is saved to the motor, and the scope never enables or moves it. **Clear E-stop** clears the motor's EStopped alert when it is set.

12.9 When Something Is Wrong

- **No ClearPath found**: check the motor's USB cable and power; **Refresh**.
- **No reply to set-address**: ClearPath-MSP (or another program) still has the port open.
- **Fewer than 845 samples/s** in the status line: another program is busy on the USB, or the cable is poor.
- The scope disconnects after five failed reads in a row; connect again.

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Chapter 13

Keyboard and Mouse

13.1 Anywhere in the Main Window

Keys	Does
F1	this guide
Ctrl + N / Ctrl + Shift + N	new project / guided setup
Ctrl + R	generate the project report (PDF)
Ctrl + O, Ctrl + S, Ctrl + Shift + S (Save as)	open, save
Ctrl + Z / Ctrl + Y	undo / redo
Space	Run: turn the crank / play the timeline in the app
Ctrl + PgUp / Ctrl + PgDn	previous / next axis
Ctrl + L	lock / unlock the profile
Ctrl + T	Timing view for crank axes (a project with crank and timeline axes)
F9	show / hide the project tree
Shift + F9	show / hide the Cam or Timing view
Ctrl + Shift + O	ClearPath scope

13.2 Displacement Diagram

Do	To
drag a knot	move it (Shift: up / down only, Ctrl: sideways only, Alt: no snap)
double-click a segment	split it there
right-click a segment	its law, or delete it
click / drag empty space	scrub the master
Ctrl +wheel, Ctrl + / −	zoom the master axis
Ctrl + 0, **Fit**	the whole cycle
middle-drag, Shift +wheel	pan while zoomed
Outputs band: double-click / drag / right-click	add / move or resize / delete a pulse

13.3 Cam View

drag	turn the cam
wheel (Shift +wheel)	step 1° (10°)

13.4 ClearPath Scope

F1	the scope chapter
Ctrl +wheel, Ctrl + / −	zoom the timebase
middle-drag, Shift +wheel	pan through time
Ctrl + 0	fit everything recorded
hover	read every lane at that time

13.5 This Guide

Enter / Shift + Enter in Find	next / previous match
Print chapter..., **Print all...**, **Save PDF...**	on paper or as a PDF

CHAPTER 14

Troubleshooting and Safety

14.1 Safety First

- The Workbench and its firmware are **experimental**: not certified for production or safety-critical use.
- Keep a **physical emergency stop** in the motor power circuit, and wire an estop input (A-9 unless you pick another) (Board page, Estop (section 9.5)).
- Run every new profile in **Simulate** first, then with **Hand jog**, then slowly.
- Keep motor speeds well inside the motors' ratings (Analysis shows them).

14.2 The Profile

You see	Do
Red warning sign at a knot, *velocity step*	the velocity jumps there (a hard knock): use a law that ends at rest, or a Blend next to a Linear segment
Amber warning sign at a knot, *acceleration step*	infinite jerk: use Cycloidal, a polynomial or modified laws instead of Harmonic / Linear
Peak velocity / acceleration > limit	slow the master, lengthen the move, or pick a gentler law
Pressure angle > 30°, undercut	cam drawing only: a larger base radius, a smaller roller
M-n is assigned to ... and ...	two axes (or the crank motor) on one motor connector

14.3 The Board

You see	Do
Amber *firmware ..., this app needs ...*	the board and the app don't match: update the firmware (section 9.13) from the Board page
Discover finds nothing	check the network cable and the PC's address range, or connect over USB: the Network (section 9.7) settings on the Board page show the board's address and can change it
Enable fails, *HLFB*	the motors did not report ready: check their power and that the profile's connectors have motors
Status: *profile is not this project*	Upload again (any change, even to the drawing, changes the hash)
Status: *not kept in flash*	the profile is lost at reboot: Upload again
Encoder counts down	tick **Reverse encoder direction** and Upload; do not swap wires
Trigger greyed	its tooltip says why: Start first, or the timeline is running
Cut length refused	below the shortest piece (the Crank page shows it)

You see	Do
Profile gone after a firmware update	the update puts it back; flashing any other way erases it: Upload again
Start refused: … *off its cam position*	Axis jog (or a stop part-way through a cycle) left an axis away from where the cam has it: choose **Move to cam positions** in the window that opens, or move it in Axis jog (section 9.4), then Start
Start refused: … *isn't homed*	the axis has **Required before Start**: home it (section 6.7) (Home all), then Start
home failed on X: the switch wasn't found within the search length	the switch didn't trip: watch its State on the Inputs page while you move the axis onto it by hand or with Axis jog; check **Toward**, and that the search length covers the travel
home failed on X: a limit tripped first	the search went the wrong way, or the home switch is beyond the limit: check **Toward**; the switch wizard proves which side it is on
The axis runs well past the home switch	set the homing **Decel** (or the axis's Acc max), or a lower search speed
The timeline stops and says *Waiting for …*	a wait (section 3.3) holds it: its input isn't at the level it wants. Check the input's State on the Inputs page; with **Change** ticked it must also go to the other level first. A PLC-bit wait needs the PLC connected and in Run
wait 1 (…) timed out	the input didn't come within the wait's timeout: Fault reset / Clear fault, find why the sensor didn't change, Start again (the axes may need **To cam position**)
A panel button does nothing	**Operator inputs…** shows each input's state live: check it reads *Pressed* when pushed (NC or NO set right?). Enable, Fault reset and Start are ignored while the PLC has control, and act only when pressed, not when already held at power-up
Upload refused: … *is the operator … input*	the project uses a pin the board's operator inputs have: move one or the other (Board, Panel)

CHAPTER 15

PLC (EtherNet/IP)

The ClearCore is also an **EtherNet/IP adapter**: a PLC reads the machine's state every few milliseconds and, when it takes control, enables, starts, stops, triggers and jogs it, and sets its speed, timeline override and cut length. The profile itself still comes from this app (**Upload**); the PLC runs it.

15.1 Adding the Board to a PLC Project

Add the board as a generic EtherNet/IP module with these values:

Setting	Value
Identity	Vendor 65000, device type 43 (generic device), product code 3, "ECam ClearCore"
Output (PLC → board)	assembly **150**, **64 bytes** (16 DINT), with a run/idle header
Input (board → PLC)	assembly **100**, **160 bytes** (40 DINT)
Configuration	assembly **151**, 0 bytes
RPI	10–100 ms; **20 ms** is a good start
Connection	Exclusive owner; input point-to-point or multicast. One owner at a time.

- **Logix** (Studio 5000): a *Generic Ethernet Module*, Comm Format *Data – DINT*, Input instance 100 size 40, Output 150 size 16, Configuration 151 size 0, the board's IP address.
- **Micro800** (CCW): a *Generic EtherNet/IP device* under the controller's Ethernet settings, with the board's IP address, an Exclusive Owner connection, input instance 100, output 150 and configuration 151 (0 bytes). Pick **DINT** for the data and the sizes are 40 (input) and 16 (output): the variables then match the maps below as they are. With **SINT** they are 160 and 64 bytes; copy the input into a `DINT[40]` and a `DINT[16]` into the output each scan with COP. The configuration variable comes out larger than set: it carries nothing, leave it.

> **NOTE: Connection timeout:** uploading a profile pauses the board's EtherNet/IP traffic about **50 ms** (the flash is being written). Keep the timeout (RPI × multiplier) at **80 ms or more**: 20 ms × 4, or 10 ms × 8. With 10 ms × 4 the PLC would see the connection drop during an upload.

> **NOTE: Wrong sizes** show as a module fault 16#0127 (output size) or 16#0128 (input size): check the sizes above. 16#0106: another PLC (or tool) already owns the connection.

15.2 Who Is in Control

The PLC is **in control** while its connection is open, it is in **Run** and it holds **PLC_CONTROL** (ControlFlags bit 1). The board strip then shows *PLC … in control* with an amber light and greys out Upload, Forget, Enable, Simulate, Start, Trigger, Hand jog and the cut length: the board refuses

them. **Stop, Abort and Disable always work**, from the app or the PLC. Without PLC_CONTROL a PLC only reads, and can still stop, abort and drop ENABLE. Releasing PLC_CONTROL hands the running machine back to the app without stopping it.

- **Edges** (START, STOP, ABORT, CLEAR_FAULT, TRIGGER) act when a bit turns on. Hold a bit for at least one RPI. A bit already on when the connection opens, or when the PLC goes to Run, is *not* an edge: nothing starts by itself after a reconnect. Turn it off and on again.
- **Levels:** ENABLE on enables, off disables. SIMULATE on is *sim 1*, off *sim 0*. JOG on is hand mode (below).
- **Nothing to acknowledge.** A refused edge or setpoint changes nothing and shows its reason in **RejectCode**; the next accepted action clears it.

15.3 When the Connection Drops

If the connection closes, times out (a pulled cable) or the PLC goes to Program while the PLC is in control, the board makes a **controlled stop** (a driven crank and the timeline ramp down at once, as a second Stop), lets the motors settle (at most half a second), then disables them and latches **FaultCode 4** (COMM_LOSS). CLEAR_FAULT, then ENABLE and START, run it again.

CAUTION

Encoder master: the board can't ramp a line or a shaft it doesn't drive, so its followers let go at once while the line may keep moving. The PLC's own connection-loss handling must stop the line (on a flying saw, the line and the saw).

15.4 Output: The Command Assembly (150)

DINT	Bytes	Field
0	0–1	**ControlFlags** (bits below; bytes 2–3 are 0)
1	4	**CrankRpm**, 0.01 rpm, driven crank. 0 = the profile's speed
2	8	**TimelineOverride**, 0.01 % (10000 = as drawn). 0 = the profile's
3	12	**CutLength**, µm of finished piece (cut-length mode). 0 = keep the current length
4	16	**JogCrankTarget**, millidegrees since the load: rev × 360000 + angle
5	20	**JogTimeTarget**, µs of timeline time
6	24	**WaitBits**: bit n = PLC wait bit n, for the timeline's waits (section 3.3) on a PLC bit. Levels; no PLC_CONTROL needed
7–15	28–63	reserved, 0

Bit	Name	Does
0	ENABLE	on: enable the profile's motors; off: disable (always)
1	PLC_CONTROL	the PLC takes control
2	START	edge: crank axes engage (a driven crank ramps up), the timeline arms
3	STOP	edge (always): a timeline finishes its cycle, then rests; a second edge ramps it to rest at once. A crank ramps to rest

Bit	Name	Does
4	ABORT	edge: stop at once (always)
5	CLEAR_FAULT	edge: clear a latched fault
6	SIMULATE	on: run without stepping the motors or switching outputs
7	JOG	on: hand mode, the master follows the jog targets; off: comes to rest and releases (always)
8	TRIGGER	edge: starts the armed timeline, for a project whose trigger is *PLC* (*EtherNet/IP TRIGGER bit*)
9	HOME	edge: home every axis with a home switch, in order (Homing (section 6.7)); STOP stops it

- **Setpoints** apply when they change, and the non-zero ones again when the PLC takes control. Setting CrankRpm or TimelineOverride back to 0 returns to the project's value. A cut length applies from the next cut (at once with the crank off).
- **Jog:** first copy where the master is into the targets (CrankRev × 360000 + CrankAngle; TimelineTime), then turn JOG on, then move the targets. The master goes there at the project's speed and ramp, every axis following its profile.
- **Trigger timing:** TRIGGER reaches the timeline about 1 ms after the packet arrives, plus the PLC's RPI and scan. Where the trigger timing matters, use a wired trigger input (200 µs).
- **Waits:** a timeline wait on a PLC bit passes when WaitBits has that bit at the level it wants, seen in the next packet: the RPI plus the PLC's scan, 20 ms or more. Without a connection in Run it never passes. TimelineState 6, StatusFlags bit 14, WaitIndex and WaitedMs say which wait holds and for how long, so the PLC can raise its own alarm for a wait with no timeout.

15.5 Input: The Status Assembly (100)

DINT	Field (low word / high word)
0	FirmwareRev (major × 256 + minor) / **StatusFlags**
1	**FaultCode** / **RejectCode**
2	EngineState / CrankState
3	TimelineState / Outputs (bit n = IO-n)
4	ProfileHash: the loaded project (the app's status line shows the same number)
5	Heartbeat, ms since boot: if it stops changing, the board has stalled
6, 7, 8	CrankAngle (millidegrees in the cycle), CrankRev, CrankRpm (0.01 rpm)
9, 10, 11	TimelineTime (µs), TimelineCycle, TimelineOverride (0.01 %)
12	EncoderCount
13	AxisCount / Inputs (bit 0 the trigger input, bit 1 estop OK)
14–18	Cut-length mode, µm: CutLength, CutLengthNext, CutLengthMin, CutGap, SyncLeft (−1 outside the sync)
19	CommLossCount
20, 21	WaitIndex (the timeline wait holding, from 1; 0 none) / reserved, WaitedMs (how long it has held)

DINT	Field (low word / high word)
24 + 4·i	Axis i: CommandedCounts, CommandedTravel (milli-units since the load), AxisFlags / Connector

Value	StatusFlags bit	EngineState	CrankState	TimelineState
0	profile loaded	empty	off	idle
1	stored in flash	disabled	following	armed
2	enabled	ready	running	running
3	ready	armed	stopping	stopping
4	running	running	jogging	done
5	timeline armed	jogging		jogging
6	timeline done	fault		waiting
7–14	stopping, jogging, faulted, estop open, simulate, PLC in control, rejected, waiting			

AxisFlags: bit 0 used, 1 on the timeline, 2 enabled, 3 HLFB asserted, 4 motor fault, 5 alerts, 6 electronic gear, 7 on its + limit, 8 on its − limit, 9 homed, 10 homing.

15.6 Fault and Reject Codes

FaultCode	Latched until CLEAR_FAULT
1	the estop opened
2	a motor faulted (AxisFlags say which)
3	a motor's HLFB did not assert on enable
4	the connection dropped while the PLC had control
5	a limit stopped the cam (the motors stay enabled)
6	homing failed (the motors stay enabled)
7	a timeline wait timed out: stopped at it (the motors stay enabled)

RejectCode	Why nothing happened
10	no profile on the board
11	needs PLC_CONTROL
12	motors disabled (ENABLE, or SIMULATE)
13	busy: moving, or an upload in progress
14	out of range (negative, or a jog target)
15	CrankRpm, but the master is an encoder
16	CutLength, but the project is not in cut-length mode
17	CutLength shorter than CutLengthMin
18	over the limit: rpm above the crank motor's max, override above 1000 %
19	a fault is latched (CLEAR_FAULT first)
20	estop open
21	TRIGGER, but the project's trigger is not the PLC

RejectCode	Why nothing happened
22	START, but an axis is not on its cam position: moved with Axis jog (section 9.4), or a timeline axis left part-way through the cycle (a second STOP, ABORT, a fault, a lost connection). JOG the timeline to 0, then START
23	START or TRIGGER, but an axis that requires homing isn't homed (HOME first)
24	START, TRIGGER or JOG with an axis on a limit (jog it off first)

15.7 Network Settings from EtherNet/IP Tools

The board's address, DHCP / static and host name can be set from any EtherNet/IP tool (the TCP/IP Interface object), as well as with `set` on the console. A new address takes effect after the board restarts by itself; the host name at once. Both are refused while the motors are enabled or anything moves. Identity Reset type 1 puts the network back to DHCP (192.168.0.110 when no DHCP server answers).

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Glossary

Advance
The change of position of an open profile in one cycle. An index feed uses it.

Axis
One servo motor and its mechanics. It follows the master. Also called a follower.

Blend
A motion law that starts and ends at the velocities of its neighbours. It joins a dwell to a linear segment, for example.

Cam position
The position that the profile gives for an axis at the current master position. An axis must be at its cam position before a start.

Carry
Steps that the library keeps for the next sample, because a sample can send at most 100 steps or because the direction changes.

Class 1 connection
An EtherNet/IP I/O connection that sends assembly data at a fixed interval. A PLC uses it to control the ClearCore.

Closed profile
A profile that ends at its start position.

Count
One step of a motor. Motor positions are in counts.

Crank
A master that turns once for each cycle: an encoder on the machine, or a motor that the ClearCore turns (a driven crank).

Cubic spline
A curve of cubic polynomials through a list of knots, with continuous position, velocity, and acceleration.

Cut-length mode
A mode of a crank master for a flying saw. The cycle stretches to the length of the piece.

Cyclic axis
An axis whose profile repeats: each axis on a crank, and an axis on a repeating timeline that spans the full cycle.

Dwell
A segment in which the axis does not move.

Electronic gear
An axis that follows the crank at a fixed ratio, calculated in whole numbers.

Engage
To start to follow the master. An axis engages without a jump.

HLFB
High-Level Feedback. An output of a ClearPath servo that shows that the servo is enabled and has no fault.

Jerk
The rate of change of the acceleration.

Knot
The end of a segment. Knot 0 is the start of the profile.

Law
The shape of the motion in a segment, defined on the unit square.

Master
The source of the progress of the machine cycle: a crank or a timeline.

Master unit
The unit in which the firmware counts the crank: one count of the encoder or crank motor divided by the denominator of the crank gear.

Open profile
A profile that ends at a position different from its start. It advances each cycle.

Peak coefficient
The largest value of the velocity (C_v), acceleration (C_a), or jerk (C_j) of a law on the unit square.

Pitch curve
The path of the center of the roller of a plate cam.

PLC
Programmable logic controller.

Pressure angle
The angle between the direction of motion of the follower and the normal to the cam at the contact point.

Profile
The position of an axis as a function of the master position.

Q16, Q24
Fixed-point numbers: a value times 2^{16} or 2^{24}, kept as a whole number.

Sample
One period of the 5 kHz motion loop: 200 µs.

Segment
A part of a profile between two knots, with one law.

Simulation
A mode in which the master runs and the positions change, but the motors receive no steps.

Sync
The part of a flying-saw cycle in which the carriage moves at the speed of the material.

Timeline
A master that is a clock. A trigger starts its cycle.

Undercut
A part of a plate cam that the roller cannot follow, because the radius of curvature of the pitch curve is smaller than the radius of the roller.

Wait
A point on a timeline where the clock stops until a condition is true.